Ref. Ares(2023)2360340 - 31/03/2023

European Commission

Annual activity report 2022

annexes

DIRECTORATE-GENERAL FOR DEFENCE INDUSTRY AND SPACE

Table of Contents

ANNEX 1: Statement of the Director(s) in charge of Risk Management and Internal Control

"I declare that in accordance with the Commission's communication on the internal control framework [1]*, I have reported my advice and recommendations on the overall state of internal control in the DG to the Director-General.*

I hereby certify that the information provided in section 2 of the present Annual activity report and in its annexes is, to the best of my knowledge, accurate and complete."

Brussels, 29/03/2023

e-signed

Christine Bernot

Head of Unit Financial management and Internal Control coordinator of DG DEFIS

---

(1) C(2017)2373 of 19.04.2017.

ANNEX 2: Performance tables

General objective 1: A European Green Deal				
Impact indicator 1: Greenhouse gas emissions **Source of the data:** European Environmental Agency (Eurostat online datacode: sdg_13_10)				
Baseline (2018)	**Interim Milestone**		**Target (2030)**	**Latest known results (2021)**
	(2020)	(year)		
-23%	-20%	...	-55% of net GHG emissions	-30%
Impact indicator 3: Climate mainstreaming in the European Union budget **Source of the data:** European Commission Draft Budget Reports				
Baseline (2014-2020, previous MFF)[2]	**Interim Milestone**		**Target (2021-2027)**	**Latest known results (2022)**
	(2022)	(year)		
21%	30%	...	30%	34.8%[3]

Specific objective 1.1: Reliable data and services of the EU Space Programme are cornerstones for the monitoring of, and transition to climate-neutrality and ecological sustainability *Related to spending programme(s):* *EU Space Programme*				
Result indicator: Number of users of the Copernicus Climate Change Service **Source of the data:** ECMWF, https://climate.copernicus.eu/				
Baseline 2019)[4]	**Interim Milestone (please add as many columns as the number of defined milestones)**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
28 000	70 000	...	80 000	150.000

(2) Change from yearly basis to duration basis.
(3) Estimate based on commitment appropriations.
(4) The Copernicus Climate Change Service started operations in June 2018. The baseline is based on the number of registered users available by 2019.

Output indicator: Number of EGNOS-based precision approach procedures published (both APV-I and LPV-200) [5] **Source of the data:** European Satellite Services Provider				
Baseline (2020)[6]	**Interim Milestone (please add as many columns as the number of defined milestones)**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
562	Increase	...	1 150 Figure approx. corresponds to the total number of instrumental runway ends in EU Member States, including in Norway and Switzerland	870

Main outputs in 2022: **External communication actions**			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Promotion of EU space based opportunities for agriculture: "International Agriculture Show, Paris (FR)"	- Number of events organised - Number of participants	- Stands and training sessions at 1 agriculture related event in the EU - 15.000 participants	Visit of Commissioner Wojciechowski, French Ministers, State Secretaries and MEP Christophe Grudler on the booth. 30 Quizz sessions for kids organised with DG AGRI 502 757 visitors in total
Touring of the "Copernicus Earth Art" Exhibition	- Number of locations Number of participants	- 6 locations in different Member States 10.000 participants	- 5 locations (Breydel premises & Open doors days in Brussels, ESE & University in Praha, University in Brno) - 7985 participants

(5) This indicator is also be reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union.

(6) Please note that the baseline value for some DEFIS indicators which have their legal base under the EU Space Regulation and European Defence Fund proposals or which were set out in the DEFIS Strategic Plan (2020-2024) with the creation of the DG DEFIS in January 2020, is set at 2020. In such cases, the earliest known values from year 2020 are set as the baselines and the latest available results by 2020 are reported, by end of 2020. Whereas, indicators which were developed under the previous MFFs and under existing programmes which are now part of the DEFIS portfolio's, may present baseline values which started earlier than in 2020.

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Touring of the “Space for Our Planet” Exhibition	- Number of locations Number of participants	- 10 different locations in different countries - 50.000 participants	- 6 locations (Bonn, Berlin, Trento, Barcelona, Rome, New York) - 251,000 participants
Organisation of a high-level conference on the future of Copernicus, in Toulouse, (FR) 16-17 Feb. 2022)	Number of participants	1.500 participants (500 onsite + 1.000 online)	- 5,810 unique viewers and 20,930 total plays (event online only)
Promotion of EU space based opportunities at the “EU Green Week” through a stand and a side session	Number of participants	500 participants	– 125 live participants (Twitter event)
Promotion of EU space based opportunities at the UN COP27 through a side session	Number of participants	100 participants	– 396 participants (including in person attendance and unique viewers)
Organisation of the launch event of the alliance on zero emission aviation	Number of participants	500 participants	– 560 onsite participants + 13,000 video unique views of the replays
Animation of a webpage dedicated to DEFIS policies contribution to the green deal objective	Number of visits	5.000 visits	– 16,500 visits
Other important outputs			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
ESTAT SDG annual Report	EUROSTAT SDG indicators based on Copernicus for SDG 15, 14	Q3 2022	Published on 23 May 2022
Copernicus European State of the Climate Report	Adoption by the Commission	Q2 2022	The 2021 Report was published on 2 November 2022. The 2022 Report will be published in 2023. Initial data from January 2023 states that Copernicus: 2022 was a year of climate extremes, with record high temperatures and rising concentrations of greenhouse gases.

Output	Indicator	Target	Latest known results (situation on 31/12/2022)
Annual Ocean State Report	Publication in the Journal of operational oceanography	Q4 2022	The Copernicus Ocean State Report issue 6 was published on 29 September 2022

General objective 2: A Europe fit for the digital age

Impact indicator 8: Uptake of space products from the European Space Programme[7]
This indicator is measured in two parts: (1) the percentage of the EU population that have a Galileo enabled device and (2) the number of registered users downloading Copernicus data and information.
Source of the data: European Union Agency for the Space Programme (EUSPA); European Space Agency (ESA); European Organisation for the Exploitation of Meteorological Satellites; Copernicus Service entrusted entities

Baseline (2019)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
Share of population with Galileo enabled device: 35%[8] (estimate)	increase	...	increase	66%
Number of registered users downloading Copernicus data and information: 300 000[9]	increase		increase	950 000

([7]) EU28 data. EU27 data not available.
(8) EU 27 data.
([9]) Worldwide users registered on European Copernicus data access portals.

Specific objective 2.1: Modern and well-functioning EU space-enabled services to support the Union's priorities *Related to spending programme(s): EU Space programme*				
Output indicator: Reliability, availability and continuity, of Copernicus Services and data stream[10] **Source of the data:** European Space Agency				
Baseline (2019)	**Interim Milestone (please add as many columns as the number of defined milestones)**		**Target (2024; less than 44 hours of interruption per month of each Sentinel mission constellation operated by ESA (corresponding to 97% availability)**	**Latest known results (2022)**
	(2022)	(year)		
Availability[11]: 94%		…	97% (tbc)	over 99.97% availability of the Space data reached
Reliability[12] (tbd)	95% (tbd)		97 % (tbc)	No figure for reliability has yet been defined
Continuity[13]: (tbc)	95% (tbc)		97% tbc)	No figure for continuity has yet been defined

(10) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union

(11) Availability in percentage of time that Copernicus Information and Data products are accessible (tbc)

(12) Reliability in one minus percentage of Copernicus Information and Data products generated that are non-conformant to the Technical Specification (tbc)

(13) Continuity in one minus percentage of Copernicus Information and Data deliveries for which the delay in becoming accessible exceeds the periodicity of that Copernicus Information and Data (tbc).

Result indicator: Availability, accuracy, and continuity of services provided by Galileo and EGNOS separately[14] **Source of the data:** Galileo and EGNOS programmes				
Baseline (2020)	**Interim Milestone** (2022)	(year)	**Target (2024)**	**Latest known results (2022)**
Galileo availability: 77%	95%		Galileo availability: 99.5%	Better than 99.5% at all user locations and for all frequency combinations
Galileo accuracy: • Horizontal positioning accuracy <= 7.5m (95%), • Vertical positioning accuracy <= 15m (95%)	Galileo accuracy: • Horizontal positioning accuracy <= 5m (95%) • Vertical positioning accuracy <= 8m (95%)		Galileo accuracy: • Horizontal positioning accuracy <= 4m • Vertical positioning accuracy <= 8m	Galileo accuracy: • Horizontal accuracy = 1.80m (95%) • Vertical accuracy = 2.80m (95%)
Galileo continuity: Not presently defined	…		Galileo continuity: Will be defined in the applicable issue of the OS SDD	No figure for continuity has yet been defined
EGNOS availability: • APV-I[15]: Over 99% of the EU territories with more than 99% of availability. • LPV-200[16]: over 90% of the EU territories with more than 99% of availability	…		EGNOS availability: • APV-I: Over 99% of the EU territories with more than 99% of availability. • LPV-200: over 95% of the EU territories with more than 99% of availability.	EGNOS availability: • APV-I: 97.4% of the EU territories with more than 99% of availability. • LPV-200: over 95% of the EU territories with more than 99% of availability

([14]) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union
([15]) APV-I = Approach procedure with vertical guidance, category 1.
([16]) LPV-200: Localizer Performance with Vertical guidance to a decision altitude of 200ft.

Baseline (2020)	Interim Milestone		Target (2024)	Latest known results (2022)
	(2022)	(year)		
EGNOS accuracy: • Horizontal (95%): 2 m • Vertical (95%): 3 m	…		EGNOS accuracy: • Horizontal (95%): 1.5 m • Vertical (95%): 2.5 m	EGNOS accuracy: • Horizontal (95%): 1.2 m • Vertical (95%): 1.7 m
EGNOS continuity: • APV-I: over 90% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 80% of the EU territories with better continuity than 5*10-4 / 15 sec.	…		EGNOS continuity: • APV-I: over 95% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 85% of the EU territories with better continuity than 5*10-4 / 15 sec.	EGNOS continuity: • APV-I: over 97% of the EU territories with better continuity than 5*10-4 / 15 sec. • LPV-200: over 88.8% of the EU territories with better continuity than 5*10-4 / 15 sec.

Result indicator: Availability of GOVSATCOM services[17] **Source of the data:** EUSPA annual market report on GOVSATCOM				
Baseline (2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
None. Testing phase, not yet fully operational		…	Number of countries participating: 27 Capacity requested: TBD	None. Testing phase, not yet fully operational

(17) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union

Main outputs in 2022:			
New policy initiatives			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Proposal for a legal basis for the establishment of an EU space-based global secure communication system adopted by the Commission	Adoption by the Commission	Q1 2022	Proposal for a Regulation establishing the new Union Secure Connectivity Programme, COM(2022) 57, of 15 February 2022
External communication actions			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Organisation of communication expert group to coordinate activities implemented by partners of the EU Space Programme	– Number of meetings – Number of factsheets Number of detailed reports	– 5 meetings – 4 quarterly factsheets – 1 annual analytical report	– 5 meetings – 4 quarterly reports – 1 annual analytical report
Organisation of a launch event for the secure connectivity initiative	Number of participants	700 participants	– Postponed to 2023
Other important outputs			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Commission implementing decision on adoption of 2022 work programmes of the Union Space Programme	Adoption by the Commission	Q1 2022	Adopted on 18. February 2022
The programme committee as foreseen in the EU space Regulation shall meet in 6 different configurations (Horizontal, Copernicus, Galileo and EGNOS, GOVSATCOM, SSA, and Security) and will assist the Commission in the implementation of the Union Space Programme	Number of meetings	3-5 per configuration in 2022	Target achieved for all formats
Launching study on space value chains resilience	Launching the study	Q1 2022	Launched in November 2022

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Launching study on improving access to testing facilities for EU new space transportation solutions, including micro-launchers	Launching the study	Q1 2022	Launched in September 2022. Final report expected in June 2023
Launching study aiming to develop a methodology for assessing the impact of EU space industry on the EU economy	Launching the study	Q1 2022	Dropped due to overlapping with another study. Budget shifted to another study
Launching study to assess extra-contractual liability risks of the EU for space activities carried out under the Union Space Programme	Launching the study	Q1 2022	Launched in Q2 2022 Study will be ready Q4 2023

Specific objective 2.2: EU Space Programme maximises socio-economic benefits *Related to spending programme(s): EU Space programme*				
Impact indicator: Market share of the European companies in the overall global GNSS market (for receivers)[18] **Source of the data:** European Union Agency for the Space Programme (EUSPA), Market report				
Baseline (2021)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2021)	(year)		
64%	…	increase	70%	64%
Impact indicator: Share of SMEs established in the EU as a proportion of the total value of the contracts relating to the Space Programme[19] **Source of the data:** DG DEFIS, European Union Agency for the Space Programme (EUSPA, European Space Agency accounting department				
Baseline (2021)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
0	increase	…	increase	No data available yet.

(18) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union

(19) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union

Result indicator: Number of start-ups supported by EU space programmes, including CASSINI initiative and R&D actions **Source of the data:** Contractors organising the activities and participating partners				
Baseline (2020)	**Interim Milestone**		**Target (2024)** The target was set by estimating the impact of the upcoming actions for startup and scaleup companies under CASSINI, notably CASSINI Business Accelerator, CASSINI Matchmaking, CASSINI Prizes and CASSINI Seed and Growth Funding Facility	**Latest known results (2022)**
	(2022)	(year)		
0	22	...	60	Above 100 (out of which 30 under Horizon)

Main outputs in 2022: **External communication actions**			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Promotion of EU space-based opportunities at space events: – 14th European Space Conference – Munich Satellite Navigation Conference – ESA living Planet Symposium – ESA Phi Week – New Space Europe Conference	– Number of events – Number of participants	– 5 events – 20.000 participants	- 3 events (since ESA Phi Week cancelled and no participation at this edition of the New Space Europe Conference) - 8.500 participants
Promotion of EU space-based opportunities at tech-events: – Hannover Messe – VivaTech – Intergeo – WebSummit Slush	– Number of events – Number of participants	– 5 events – 100.000 participants	- 4 events (no participation at Hannover Messe) - 188,000 participants

Output	Indicator	Target	Latest known results (situation on 31/12/2022)
Organisation of EU Space Week	– Number of participants	– 1.500 participants	- 3,000+ registered participants, 1,100+ onsite attendees, 18,800+ online watchers (incl. 3,350+ unique visitors)
Animation of a webpage dedicated to EU space contribution to the digital age and to CASSINI	– Number of visits	– 15.000 visits	- 25,000 visits

Other important outputs

Output	Indicator	Target	Latest known results (situation on 31/12/2022)
CASSINI space hackathon events	- Number of CASSINI Hackathon events carried out physically/online; - number of teams supported	- 2 hackathon events in 2022 - 200 teams/year supported	2 hackathon events in 2022; 600 individual participants & 100 teams supported, 6 winners per year receiving 100h mentoring. 13 teams received mentoring during 2021-22.
CASSINI Maritime Challenge	- Number of prizes published and awarded	- 3 winners to be selected for the best digital applications based on space data and signals.	Prize published in 2022, selection in 2023 of 3 winners for a prize amount of €950k.
CASSINI business accelerator	- Number of participating startup companies enrolled in CASSINI Business Accelerator	- 20 participating companies per batch (in 2022). Action launched in Q1/Q2 2022.	Contract signed in 2022. 20 participating companies starting in May 2023, 40 companies in total in 2023.
CASSINI matchmaking	- Number of participating startup companies in CASSINI Matchmaking	- 100 participating companies (in 2022). Action launched in Q1 2022.	Action launched in Q1 2022. 100 participating companies, investors and corporates in 2022. Same target in 2023.
CASSINI seed and growth funding facility: Venture capital funds signing investment agreements with EIF (InvestEU)	- Number of venture capital funds selected by EIF in 2022	- 4 investment agreements (in 2022). InvestEU and CASSINI facility to be launched in Q1 2022.	InvestEU and CASSINI facility launched in Q2 2022. Target: 4 investment agreements in 2022, outcome 2 signed. Target for 2023: 4 investment agreements signed.

Output	Indicator	Target	Latest known results (situation on 31/12/2022)
CASSINI seed and growth funding facility: companies receiving investments from venture capital funds (InvestEU)	- Number of companies selected by venture capital funds and subscription agreements signed.	- 50-60 new venture capital investments in space-related companies per year (when fully operational). Stage 1 objective in 2022: first 5 investments concluded by VC funds under CASSINI fund.	50-60 new venture capital investments in space-related companies per year (when fully operational). KPIs reported by EIF with a delay of appr 1 year.
CASSINI in-orbit demonstration and validation	- Number of open calls for expression of interest in 2022	- Number of open calls for expression of interest in 2022 + number of selected projects	(IOD/IOV to be reported separately)
Call for expression of interest for IOD/IOV experiments	- Number of calls published	- 1	(IOD/IOV to be reported separately)
Obtain demographic characterisation of equality in the defence, aeronautics and space EU sectors	Final report of survey Analysis of the implementation of EU equality legislation in the defence, aeronautics and space EU sectors	Q4 2022	x
Contribute to the development of a European cyber defence policy	Adoption of new European cyber resilience Act Adoption of the joint communication of the European Policy on cyber defence (EPCD)	Q3 2022	Joint communication was adopted 10 November 2022. Implementation plan drafting on-going.
Contribute to the development of the drone strategy 2.0 (Action 9 under the action plan on synergies for civil, defence and space Industries)	Adoption by the Commission	Q4 2022	Adopted on 29 November 2022
Contribute to the European chips Act	Adoption by the Commission	Q2 2022	Council adopted its General approach on 1 December 2022. Examination of the proposal in the European Parliament ongoing
Launching of a study on semi-conductor supply chains for the benefit of the EU space domain	Launch study	Q1 2022	The evaluation of the proposals took place in February 2022. The actual kick off took place on 24 March 2022.

General objective 3: An Economy that works for people				
Impact indicator 11: Public EU and EU Member States investments in space **Source of the data:** Euroconsult report 2021 Government Space Programs - Benchmarks, Profiles and Forecasts to 2030, and EC Central Financial Information System (ABAC)				
Baseline (2018)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
EUR 10 941 million	increase	...	increase	EUR 10 006 million[20]

General objective 4: A Stronger Europe in the world				
Impact indicator 12: Value of research and development actions funded by the European Defence Fund[21] **Source of the data:** European Commission				
Baseline (2019)	**Interim Milestone**		**Target (2024)**	**Latest known results (2021)**
	(2022)	(year)		
EUR 0	increase	...	increase	EDF: EUR 1 170 million (EDIDP: EUR 803.13 million PADR: EUR 86 million)[22]

(20) Figures from 3 EU Member States (Croatia, Cyprus, Latvia) are not included.

(21) This indicator does not reflect the EU contributions in the framework of the two ongoing precursor programmes PADR and EDIDP, supporting defence research and capability development, on which the EDF is based.

(22) EDF grant agreements will be the data source for this impact indicator. Data on 2021 includes values for grants signed at the moment of reporting (end of 2022). The latest values for the precursors PADR and EDIDP are provided for information only. The value presented for the development actions is the accumulated total eligible costs of these projects, including co-funding from the Member States.

Specific objective 4.1: Fostered innovation capacity and competitiveness of the European defence industry and strengthened EU defence supply chains due to increased cross-border R&D cooperation involving in particular SMEs and mid-caps

Related to spending programme(s):

European Defence Industrial Development Programme (EDIDP),

European Defence Fund (EDF)

EU Space Programme

Output indicator: Legal entities involved in EDF[23]

Source of the data: European Commission

Baseline (2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2021)**
	(2022)	(year)		
0	increase	…	increase	EDF: 670[24] (PADR: 139 legal entities EDIDP: 406 legal entities)

Result indicator: Proportion of budget of EDF allocated to disruptive technologies

Source of the data: European Commission

Baseline (2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2021)**
	(2022)	(year)		
0	4%	…	6%	5.74%

(23) This indicator is also reported under the Programme Performances Statement for the European Defence Fund, as part of the annual draft general budget of the European Union. This indicator does not reflect on the outcome of the two ongoing precursor programmes PADR and EDIDP, supporting defence research and capability development, on which the EDF is based.

24 In total, 1111 legal entities were involved in the admissible proposals submitted under the EDF 2021 calls. This indicator reports on unique entities, which involvement in research and development actions implemented in direct and indirect management is known at the moment of reporting (end of 2022). The latest values for the precursors PADR and EDIDP are provided for information only.

Output indicator: Share of EDF contracts awarded involving collaboration with to cross-border SME and mid-caps[25] **Source of the data:** European Commission				
Baseline (2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2021)**
	(2022)	(year)		
0	increase	...	increase	7,7 / 10,4[26]

Main outputs in 2022: **External communication actions**			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Promotion of EU space-based opportunities at the Arctic Circle Conference, 13-16 October, Reykjavik (Iceland) through a stand and a side session	- Number of participants (stand) - Number of participants (side session)	- 1.000 participants (stand) - 100 participants (side session)	- 2,000 participants
Promotion of EU space-based opportunities at ION GNSS+ 2022, 19-23 September, Denver (USA), with a stand and a side session	- Number of participants	- 1.500 participants	- DEFIS participation cancelled for this edition
Promotion of EU space-based opportunities at EU Development Days, in June 2022, Brussels, (Belgium) with a stand and a side session	- Number of participants	- 2.000 participants	- 2,500 participants - 70 visits to the stand
Increase awareness of the achievements of PADR, EDIDP (the precursors of EDF) and EDF	- Number of engagement on social media	- 300.000 engagements on social media channels	- 11,026 engagements - 333,354 impressions - 84,000 video views for 4 videos - 30,000 Factsheet downloads
Organisation of EDF Info Day	- Number of participants	- 2.000 participants	- 1,600 participants remotely and on-site

(25) This indicator is also reported under the Programme Performances Statement for the European Defence Fund, as part of the annual draft general budget of the European Union.

26 Computation based on data available for 11 research projects / 14 development projects.

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Participation in national EDF Info Days throughout 2022 with expert speakers (depending on the number of events organised by Member States)	- Number of events - Number of participants per event	- 5 events - 200 participants	- 9 events - 450 participants
Organisation of thematic conferences	- Number of events - Number of participants	- 24 events - 2400 participants	- 7 events - 2,600 participants
Promotion of EU's defence industry initiatives at fairs	- Number of participants	- 2.000 attendees	- 1,000 attendees
Other important outputs			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Commission Implementing Decision on the annual work programme for the European Defence Fund 2022	Adoption by the Commission	Q2 2022	Adoption on 25 May 2022
Announcement of the EU Defence Innovation Scheme (EUDIS)	Publication on website		Publication of factsheet on 22 May 2022
Issue forth implementation Report on the action plan on military mobility (Joint Report COM/HR)	Adoption by the Commission	Q3 2022	Adoption of the Action Plan on Military Mobility 2.0 on 10 November 2022
Establishment of the European defence industry Reinforcement through common Procurement Act (Regulation)	Regulation proposal Adoption by the Commission	Q3 2022	Council adopted its General approach on 1 December 2022. Discussion in the European Parliament ongoing.

General objective 5: Promoting our European way of life				
Impact indicator 7: Victims of terrorist attacks **Source of the data:** The Europol Te-Sat report				
Baseline **(2017)**	**Interim Milestone**		**Target** **(2024)**	**Latest known results** **(2021)**
	(2022)	(year)		
32 people died as a result of terrorist attacks 157 people were injured	Decrease	…	decrease	2 died as a result of terrorist attacks in the EU. 17 people were injured.

Specific objective 5.1: Security actors have access to EU autonomous tools, space-enabled services, and technologies, needed to build resilience to security threats, safety hazards and crisis situations *Related to spending programme(s):* *EU Space Programme,* *Horizon 2020/Europe* *European Defence Fund (EDF)*				
Result indicator: Number of operational safety and security (including dual-use) related services from the EU Space Programmes **Source of the data:** European Commission (DG DEFIS), EU Space Programme				
Baseline (2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
• GALILEO Public Regulated Service (PRS) Initial Service • EGNOS Safety of Life (SoL) service • GALILEO Search and Rescue (SAR) service • Copernicus security service[27]/ emergency service • Space Surveillance and Tracking (SST)	• GALILEO Public Regulated Service (PRS) initial service • EGNOS Safety of Life (SoL) service • GALILEO Search and Rescue (SAR) service • Copernicus security service/ emergency service • Space Surveillance and Tracking (SST)	…	• GALILEO Public Regulated Service (PRS) Full Operational Capability (FOC)[28] • EGNOS Safety of Life (SoL) service • GALILEO Search and Rescue (SAR) service • GALILEO Emergency Warning Service (EWS) • Copernicus security service/ emergency service • Space Surveillance and Tracking (SST) Space Weather (SWE) • Near Earth Objects (NEO) • GOVSATCOM (Governmental Satellite Communications)	• GALILEO Public Regulated Service (PRS) initial service • EGNOS Safety of Life (SoL) service • GALILEO Search and Rescue (SAR) service with return link feature since 2020. • GALILEO Emergency Warning Service (EWS) is in the service definition stage. • Copernicus security service/ emergency service • Space Surveillance and Tracking (SST)

(27) (1) Copernicus Maritime Surveillance Service, (2) Copernicus Border Surveillance Service, (3) Copernicus Support to EU External Actions

(28) The schedule is at risk. The new target date is under assessment

Result indicator: EU Funding for critical technologies for European strategic autonomy in aerospace and defence **Source of the data:** DG DEFIS Programmes				
Baseline (2014-2020)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2022)	(year)		
EUR 105 M[29]				EUR 10.5 M (space–related activities)

Output indicator: Number of EU launches for the Space Programme (including numbers by type of launchers)[30] **Source of the data:** DG DEFIS, EU Space Programme				
Baseline (2021)	**Interim Milestone**		**Target (2024)**	**Latest known results (2022)**
	(2023)	(year)		
0	2 (1 contracted and scheduled, 1 planned)	...	3 (1 launch contracted and still to be scheduled, 2 launches planned)	1 launch in 2021 (Galileo#L11)

(29) The indicator measures the level of funding allocated from Horizon 2020/Europe to the calls dedicated to critical space technologies as defined in the critical space technologies list established by the joint task force (JTF) of the European Commission, ESA and EDA. The JTF list provides critical technologies, at component or equipment level, that are not available from a European source and for which the unrestricted availability from non-European suppliers cannot be assured. Additional funding for the development of technologies in the JTF list should lead to wider availability of such critical technologies for the EU aerospace and defence systems and accordingly to the European strategic non-dependence.

(30) This indicator is also reported under the Programme Statement for the EU Space Programme, as part of the annual draft general budget of the European Union.

Main outputs in 2022:

New policy initiatives

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Communication on Development of an EU approach on Space Traffic Management (STM)	Adoption by the Commission	Q1 2022	Joint Communication on ‘An EU Approach for Space Traffic Management - An EU contribution addressing a global challenge’ published on 15 February 2022

External communication actions

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Organisation of the launch event of the alliance on space launchers	- Number of participants	- 700 participants	Postponed
Touring Exhibition “EU Space for You”	- Number of locations Number of participants	- 5 different locations - 50.000 participants	- 4 different locations (Winter Wonders Brussels, EU Open Doors Day Brussels, Web Summit Lisbon, EU Space Week Prague)
Promotion of DEFIS policies at EU Open Doors Day	- Number of participants	- 5.000 participants	- 12.000 visitors
Partnership with Brussels city to promote EU space-based opportunities at “Winter Wonders” through interactive exhibition and information sessions	- Number of participants	- 500.000 participants	- 3,444,762,00 participants (1,236,091 from Brussels, and 1,032,015 people from elsewhere in the country) and 1,176,656 internationals.
Partnership with EU-located museums to permanently promote opportunities offered by EU Space	- Number of partnerships signed	- 2 partnerships	Postponed

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Promotion of EU space / Copernicus Emergency Management Service activation	- Number of engagement/impressions on social media	- 3.000.000 engagements/impressions on social media channels	- 10,013,153 impressions (DEFIS social media TW) - 338,772 engagements (DEFIS social media TW) - 37,593,533 impressions (Copernicus social media TW, FB, IG) - 1,747,852 engagements (Copernicus social media TW, FB, IG)
Organisation of the Galileo SAR live demonstration event	- Number of participants (physical and online) - Number of engagement/impressions on social media - Number of events - Number of journalists/influencers following the event	- 5.000 participants - 1 million engagements/impressions on social media channels - 1 live demonstration organised - 20 journalists/influencers	Postponed
Communication actions to launch the new EU Space Programme components SSA and GovSatCom	- Number of events - Number of participants - Media coverage	- 2 events - 150 participants - 50 media mentions	Postponed
Animation of a dedicated webpage on DEFIS Europa platform promoting DEFIS contribution to this objective	- Number of visits	- 5.000 visits	- 9,000 visits

Other important outputs

Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Implementing act on the procedure for the establishment of the future EU SST partnership	Adoption by the Commission	Q1 2022	COMMISSION IMPLEMENTING DECISION (EU) 2022/1245 of 15 July 2022

Output	Indicator	Target	Latest known results (situation on 31/12/2022)
Conclude negotiations with Arianespace on the Inter-Institutional framework contract for EU launch services	Adoption by the Commission (conclusion of the framework contract)	Q1 2022	Framework contract signed on 29 November 2022
Launch and set up a European alliance on space launchers	Launch European alliance on Space launchers	Q1 2022	Not launched yet, under consideration

ANNEX 3: Draft annual accounts and financial reports

TABLE 1: OUTTURN ON COMMITMENT APPROPRIATIONS IN 2022 (in Mio €) for DG DEFIS					
			Commitment appropriations authorised*	**Commitments made**	**%**
			1	2	3=2/1
Title 01 Research and Innovation					
01	01 01	Support administrative expenditure of the "Research and Innovation" cluster	3,73	1,43	38,36 %
	01 02	Horizon Europe	84,59	84,32	99,68 %
	01 20	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00 %
Total Title 01			**88,32**	**85,75**	**97,09 %**
Title 03 Single Market					
03	03 02	Single Market Programme	0,00	0,00	0,00 %
	03 20	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00 %
Total Title 03			**0,00**	**0,00**	**0,00 %**
Title 04 Space					
04	04 01	Support administrative expenditure of the 'Space' cluster	6,56	6,06	92,42 %
	04 02	Space programme of the Union	2.055,48	2.006,45	97,61 %
	04 10	Decentralised agencies	70,03	70,03	100,00 %
Total Title 04			**2.132,07**	**2.082,54**	**97,68 %**
Title 08 Agriculture and Maritime Policy					
08	08 02	European Agricultural Guarantee Fund (EAGF)	1,90	1,90	100,00 %
Total Title 08			**1,90**	**1,90**	**100,00 %**
Title 09 Environment and Climate Action					
09	09 02	Programme for the Environment and Climate Action (LIFE)	0,00	0,00	0,00 %
	09 20	Pilot projects, preparatory actions, prerogatives and other actions	0,99	0,99	100,00 %
Total Title 09			**0,99**	**0,99**	**100,00 %**
Title 13 Defence					
13	13 01	Support administrative expenditure of the "Security and Defence" cluster	1,76	1,76	100,00 %
	13 02	European Defence Fund (EDF) - Non research	639,48	639,48	100,00 %
	13 03	European Defence Fund (EDF) - Research	320,00	320,00	100,00 %
	13 20	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00 %
Total Title 13			**961,25**	**961,25**	**100,00 %**
Title 14 External Action					
14	14 02	Neighbourhood, Development and International Cooperation Instrument - Global Europe (NDICI - Global Europe)	0,00	0,00	0,00 %
Total Title 14			**0,00**	**0,00**	**0,00 %**
Title 20 Administrative expenditure of the European Commission					

20	20 02	Other staff and expenditure relating to persons	0,01	0,01	100,00 %
Total Title 20			**0,01**	**0,01**	**100,00 %**
Total Excluding NGEU			**3.184,55**	**3.132,45**	**98,36 %**

Title 01 Research and Innovation					
01	01 02	Horizon Europe	35,24	35,24	100,00 %
Total Title 01			**35,24**	**35,24**	**100,00 %**
Total NGEU Only			**35,24**	**35,24**	**100,00 %**

Total DG DEFIS	**3.219,78**	**3.167,69**	**98,38 %**

** Commitment appropriations authorised include, in addition to the budget voted by the legislative authority, appropriations carried over from the previous exercise, budget amendments as well as miscellaneous commitment appropriations for the period (e.g. internal and external assigned revenue).*

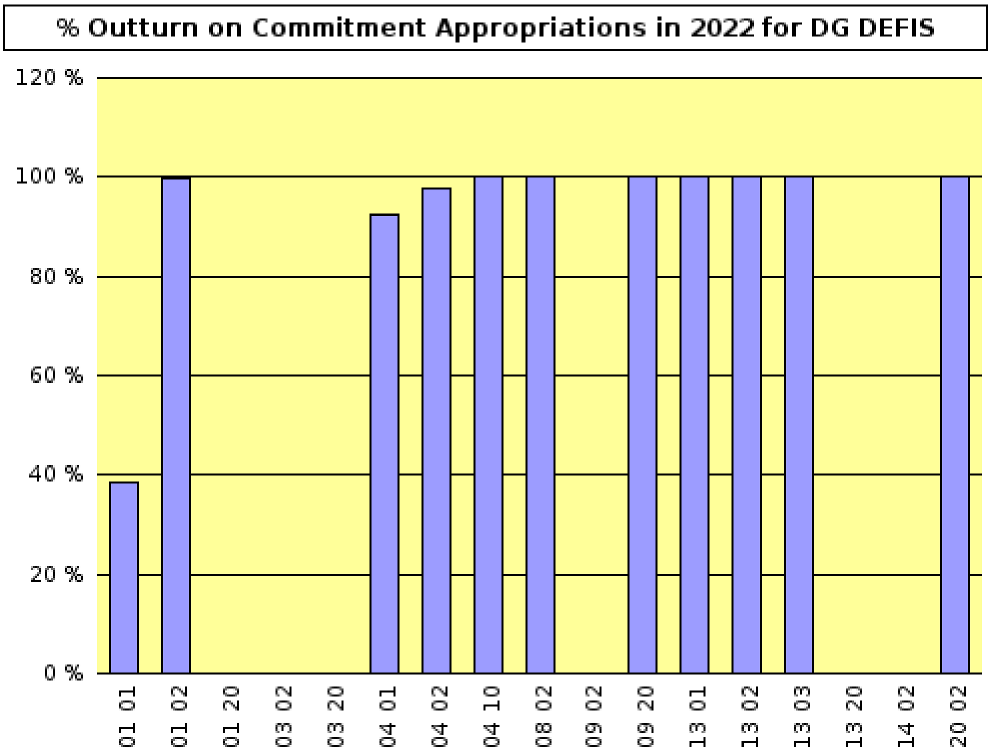


TABLE 2: OUTTURN ON PAYMENT APPROPRIATIONS in 2022 (in Mio €) for DG DEFIS			
	Payment appropriations authorised *	**Payments made**	**%**
	1	**2**	**3=2/1**
Title 01 Research and Innovation			

01	01 01	Support administrative expenditure of the "Research and Innovation" cluster	4,89	1,36	27,84 %
	01 02	Horizon Europe	83,94	47,19	56,23 %
	01 20	Pilot projects, preparatory actions, prerogatives and other actions	2,92	2,92	100,00 %
Total Title 01			**91,75**	**51,48**	**56,10%**
Title 03 Single Market					
03	03 02	Single Market Programme	0,16	0,16	100,00 %
	03 20	Pilot projects, preparatory actions, prerogatives and other actions	0,47	0,00	0,00 %
Total Title 03			**0,63**	**0,16**	**25,05%**
Title 04 Space					
04	04 01	Support administrative expenditure of the 'Space' cluster	11,06	5,56	50,25 %
	04 02	Space programme of the Union	2.171,97	1.983,82	91,34 %
	04 10	Decentralised agencies	65,39	65,39	100,00 %
Total Title 04			**2.248,42**	**2.054,78**	**91,39%**
Title 08 Agriculture and Maritime Policy					
08	08 02	European Agricultural Guarantee Fund (EAGF)	0,40	0,40	100,00 %
Total Title 08			**0,40**	**0,40**	**100,00%**
Title 09 Environment and Climate Action					
09	09 02	Programme for the Environment and Climate Action (LIFE)	0,00	0,00	0,00 %
	09 20	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00 %
Total Title 09			**0,00**	**0,00**	**0,00%**
Title 13 Defence					
13	13 01	Support administrative expenditure of the "Security and Defence" cluster	2,70	1,45	53,77 %
	13 02	European Defence Fund (EDF) - Non research	361,10	361,10	100,00 %
	13 03	European Defence Fund (EDF) - Research	158,19	158,19	100,00 %
	13 20	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00 %
Total Title 13			**521,99**	**520,74**	**99,76%**
Title 14 External Action					
14	14 02	Neighbourhood, Development and International Cooperation Instrument - Global Europe (NDICI - Global Europe)	0,00	0,00	0,00 %
Total Title 14			**0,00**	**0,00**	**0,00%**
Title 20 Administrative expenditure of the European Commission					
20	20 02	Other staff and expenditure relating to persons	0,02	0,00	11,38 %
Total Title 20			**0,02**	**0,00**	**11,38%**
Total Excluding NGEU			**2.863,22**	**2.627,56**	**91,77%**

Title 01 Research and Innovation

01	01 02	Horizon Europe	0,31	0,31	99,97 %
Total Title 01			**0,31**	**0,31**	**99,97%**
Total NGEU Only			**0,31**	**0,31**	**99,97%**

Total DG DEFIS	**2.863,53**	**2.627,86**	**91,77 %**

** Payment appropriations authorised include, in addition to the budget voted by the legislative authority, appropriations carried over from the previous exercise, budget amendments as well as miscellaneous payment appropriations for the period (e.g. internal and external assigned revenue).*

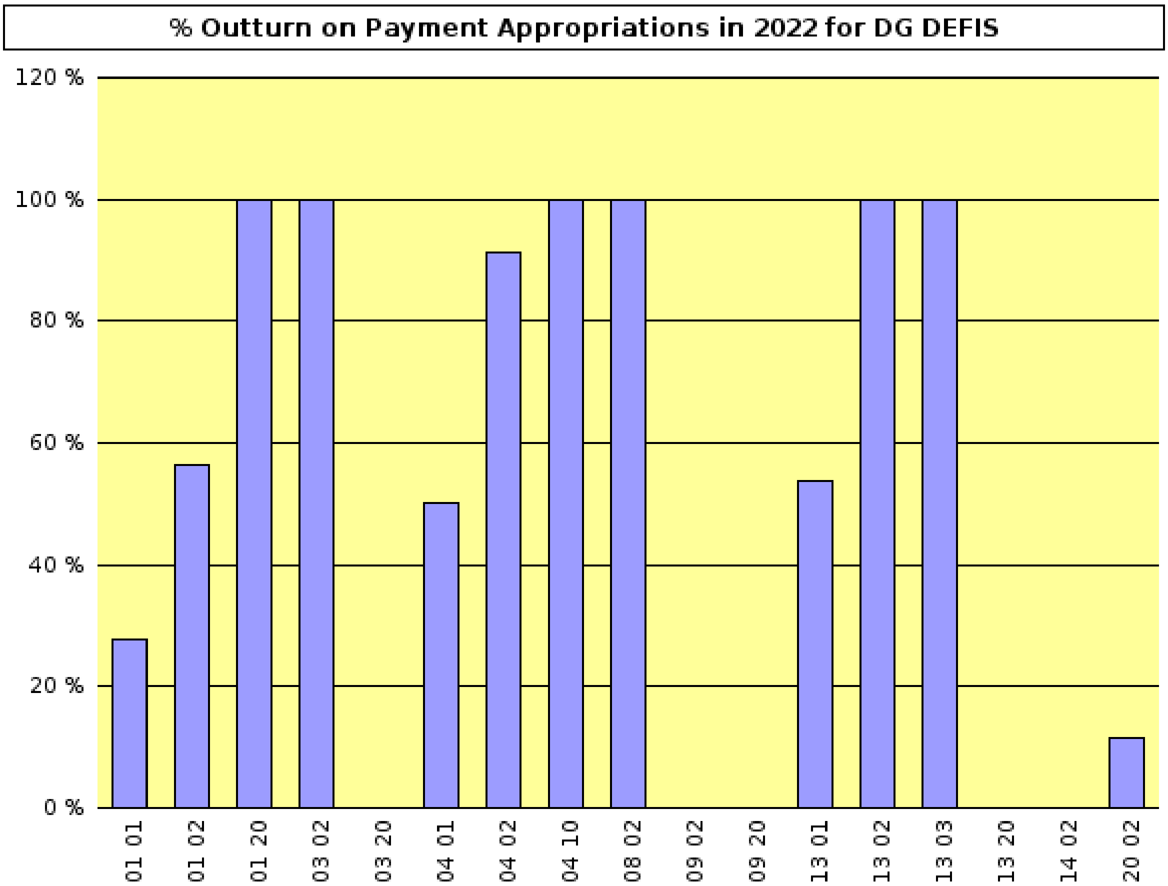


TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				**Commitments to be settled from financial years previous to 2021**	**Total of commitments to be settled at end of financial year 2022**	**Total of commitments to be settled at end of financial year 2021**
Chapter			**Commitments**	**Payments**	**RAL**	**% to be settled**			
			1	**2**	**3=1-2**	**4=1-2/1**	**5**	**6=3+5**	**7**
01	01 01	Support administrative expenditure of the "Research and Innovation" cluster	1,43	0,34	1,09	76,36%	0,00	1,09	1,17
	01 02	Horizon Europe	84,32	2,86	81,47	96,61%	137,84	219,30	182,41

	0120	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00	0,00%	2,48	2,48	5,46
Total Title 01			**85,75**	**3,19**	**82,56**	**96,28%**	**140,32**	**222,88**	**189,04**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
03	0302	Single Market Programme	0,00	0,00	0,00	0,00%	0,72	0,72	0,98
	0320	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00	0,00%	0,51	0,51	0,51
Total Title 03			**0,00**	**0,00**	**0,00**	**0,00%**	**1,22**	**1,22**	**1,49**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
04	0401	Support administrative expenditure of the 'Space' cluster	6,06	1,63	4,44	73,14%	0,00	4,44	4,54
	0402	Space programme of the Union	2.006,45	1.246,66	759,78	37,87%	600,18	1.359,97	1.339,83
	0410	Decentralised agencies	70,03	50,01	20,02	28,59%	0,00	20,02	15,38
Total Title 04			**2.082,54**	**1.298,30**	**784,24**	**37,66%**	**600,19**	**1.384,43**	**1.359,75**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
08	0802	European Agricultural Guarantee Fund (EAGF)	1,90	0,00	1,90	100,00%	0,80	2,70	1,20
Total Title 08			**1,90**	**0,00**	**1,90**	**100,00%**	**0,80**	**2,70**	**1,20**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				Commitments to be settled from financial	Total of commitments to be settled at	Total of commitments to be settled at

Chapter			Commitments	Payments	RAL	% to be settled	years previous to 2021	end of financial year 2022	end of financial year 2021
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
09	0902	Programme for the Environment and Climate Action (LIFE)	0,00	0,00	0,00	0,00%	0,90	0,90	0,90
	0920	Pilot projects, preparatory actions, prerogatives and other actions	0,99	0,00	0,99	100,00%	0,00	0,99	0,00
Total Title 09			**0,99**	**0,00**	**0,99**	**100,00%**	**0,90**	**1,89**	**0,90**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS

			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
13	1301	Support administrative expenditure of the "Security and Defence" cluster	1,76	0,55	1,21	68,73%	0,00	1,21	0,94
	1302	European Defence Fund (EDF) - Non research	639,48	72,00	567,48	88,74%	471,51	1.038,99	761,54
	1303	European Defence Fund (EDF) - Research	320,00	0,00	320,00	100,00%	170,28	490,28	328,47
	1320	Pilot projects, preparatory actions, prerogatives and other actions	0,00	0,00	0,00	0,00%	1,97	1,97	1,97
Total Title 13			**961,25**	**72,56**	**888,69**	**92,45%**	**643,75**	**1.532,45**	**1.092,91**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS

			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
14	1402	Neighbourhood, Development and International Cooperation Instrument - Global Europe (NDICI - Global Europe)	0,00	0,00	0,00	0,00%	6,73	6,73	6,73
Total Title 14			**0,00**	**0,00**	**0,00**	**0,00%**	**6,73**	**6,73**	**6,73**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS

	Commitments to be settled	Commitments to be settled from financial	Total of commitments to be settled at	Total of commitments to be settled at

Chapter			Commitments	Payments	RAL	% to be settled	years previous to 2021	end of financial year 2022	end of financial year 2021
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
20	2002	Other staff and expenditure relating to persons	0,01	0,00	0,01	83,94%	0,00	0,01	0,01
Total Title 20			**0,01**	**0,00**	**0,01**	**83,94%**	**0,00**	**0,01**	**0,01**
Total Excluding NGEU			**3.132,45**	**1.374,06**	**1.758,40**	**56,13%**	**1.393,91**	**3.152,30**	**2.652,03**

TABLE 3 : BREAKDOWN OF COMMITMENTS TO BE SETTLED AT 31/12/2022 (in Mio €) for DG DEFIS									
			Commitments to be settled				Commitments to be settled from financial years previous to 2021	Total of commitments to be settled at end of financial year 2022	Total of commitments to be settled at end of financial year 2021
Chapter			Commitments	Payments	RAL	% to be settled			
			1	2	3=1-2	4=1-2/1	5	6=3+5	7
01	0102	Horizon Europe	35,24	0,31	34,93	99,13%	9,60	44,53	9,60
Total Title 01			**35,24**	**0,31**	**34,93**	**99,13%**	**9,60**	**44,53**	**9,60**
Total NGEU Only			**35,24**	**0,31**	**34,93**	**99,13%**	**9,60**	**44,53**	**9,60**

Total for DG DEFIS	**3.167,69**	**1.374,36**	**1.793,32**	**56,61 %**	**1.403,51**	**3.196,83**	**2.661,63**

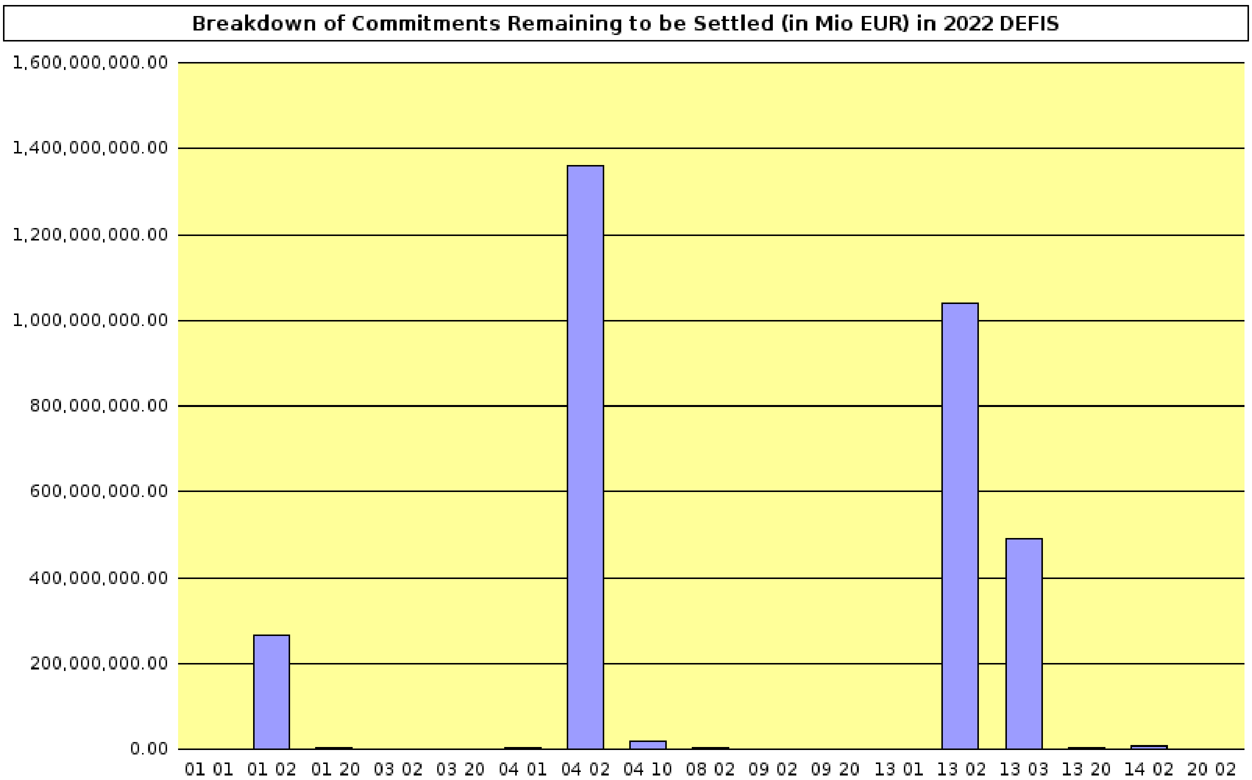


TABLE 4 : BALANCE SHEET for DG DEFIS		
BALANCE SHEET	**2022**	**2021**
A.I. NON CURRENT ASSETS	9.010.244.453,55	8.435.948.337,91
A.I.1. Intangible Assets	2.188.989,06	2.747.879,88
A.I.2. Property, Plant and Equipment	8.483.638.182,34	8.128.497.204,22
A.I.5. Non-Current Pre-Financing	524.417.282,15	304.703.253,81
A.II. CURRENT ASSETS	2.630.491.835,46	2.293.155.539,79
A.II.2. Current Pre-Financing	2.630.491.835,46	2.293.134.019,85
A.II.3. Curr Exch Receiv &Non-Ex Recoverables	0,00	21.519,94
ASSETS	**11.640.736.289,01**	**10.729.103.877,70**
P.II. CURRENT LIABILITIES	-36.632.626,30	-15.765.148,91
P.II.4. Current Payables	-717.299,24	-342.141,73
P.II.5. Current Accrued Charges &Defrd Income	-35.915.327,06	-15.423.007,18
LIABILITIES	**-36.632.626,30**	**-15.765.148,91**
NET ASSETS (ASSETS less LIABILITIES)	**11.604.103.662,71**	**10.713.338.728,79**

Non-allocated central (surplus)/deficit*	-15.588.012.046,97	-13.495.819.360,75
P.III.2. Accumulated Surplus/Deficit	3.983.908.384,26	2782480632
TOTAL DG DEFIS	**0,00**	**0,00**

It should be noted that the balance sheet and statement of financial performance presented in Annex 3 to this Annual Activity Report, represent only the assets, liabilities, expenses and revenues that are under the control of this Directorate General. Significant amounts such as own resource revenues and cash held in Commission bank accounts are not included in this Directorate General's accounts since they are managed centrally by DG Budget, on whose balance sheet and statement of financial performance they appear. Furthermore, since the accumulated result of the Commission is not split amongst the various Directorates General, it can be seen that the balance sheet presented here is not in equilibrium.

Additionally, the figures included in tables 4 and 5 are provisional since they are, at this date, still subject to audit by the Court of Auditors. It is thus possible that amounts included in these tables may have to be adjusted following this audit.

TABLE 5 : STATEMENT OF FINANCIAL PERFORMANCE for DG DEFIS

STATEMENT OF FINANCIAL PERFORMANCE	2022	2021
II.1 REVENUES	-368.246.252,15	-405.694.870,13
II.1.1. NON-EXCHANGE REVENUES	-373.659.751,51	-408.879.339,70
II.1.1.6. RECOVERY OF EXPENSES		-21.519,94
II.1.1.8. OTHER NON-EXCHANGE REVENUES	-373.659.751,51	-408.857.819,76
II.1.2. EXCHANGE REVENUES	5.413.499,36	3.184.469,57
II.1.2.1. FINANCIAL INCOME		-170,66
II.1.2.2. OTHER EXCHANGE REVENUE	5.413.499,36	3.184.640,23
II.2. EXPENSES	2.048.247.358,53	1.607.122.622,43
II.2. EXPENSES	2.048.247.358,53	1.607.122.622,43
II.2.10.OTHER EXPENSES	903.999.097,22	630.153.171,49
II.2.2. EXP IMPLEM BY COMMISS&EX.AGENC. (DM)	177.432.543,89	114.030.661,95
II.2.3. EXP IMPL BY OTH EU AGENC&BODIES (IM)	503.804.865,84	446.612.720,68
II.2.4. EXP IMPL BY 3RD CNTR & INT ORG (IM)	427.031.314,25	366.344.300,64
II.2.5. EXP IMPLEM BY OTHER ENTITIES (IM)	35.979.537,33	49.980.178,64
II.2.8. FINANCE COSTS		1.589,03
STATEMENT OF FINANCIAL PERFORMANCE	**1.680.001.106,38**	**1.201.427.752,30**

TABLE 5bis : OFF BALANCE SHEET for DG DEFIS

OFF BALANCE	2022	2021
OB.1. Contingent Assets	73.015.814,22	44.914.139,30
GR for pre-financing	73.015.814,22	44.914.139,30
OB.3. Other Significant Disclosures	-7.217.590.870,92	-7.217.590.870,92
OB.3.2. Comm against app. not yet consumed	-2.645.954.703,24	-2.645.954.703,24
OB.3.3.21. LC - Horizon EU	-382.498.121,87	-382.498.121,87
OB.3.3.4.Galileo programme	-4.189.138.045,81	-4.189.138.045,81
OB.4. Balancing Accounts	6.105.333.306,70	6.133.434.981,62
OB.4. Balancing Accounts	6.105.333.306,70	6.133.434.981,62
OFF BALANCE	**-1.039.241.750,00**	**-1.039.241.750,00**

Table 6 : AVERAGE PAYMENT TIMES in 2022 for DEFIS

Legal Times									
Maximum Payment Time (Days)	Total Nbr of Payments	Nbr of Payments within Time Limit	Percentage	Average Payment Times (Days)	Nbr of Late Payments	Percentage	Average Payment Times (Days)	Late Payments Amount	Percentage
30	430	429	99,77	11,45	1	0,23 %	36,00	23.980,	0, %

			%					00	
60	71	71	100,00 %	20,15				0,00	0, %
90	16	16	100,00 %	47,13				0,00	0, %

Total Number of Payments	**517**	**516**	**99,81 %**		**1**	**0,19 %**		**23980**	**0, %**
Average Net Payment Time	**13,80077369**			**13,76**			**36,00**		
Average Gross Payment Time	**16,98839458**			**16,95155**			**36**		

Suspensions							
Average Report Approval Suspension Days	**Average Payment Suspension Days**	**Number of Suspended Payments**	**% of Total Number**	**Total Number of Payments**	**Amount of Suspended Payments**	**% of Total Amount**	**Total Paid Amount**
0	26	64	12,38 %	517	204.368.103,10	7,78 %	2.627.883.570,86

TABLE 7 : SITUATION ON REVENUE AND INCOME in 2022 for DG DEFIS								
Chapter		**Revenue and income recognized**			**Revenue and income cashed from**			**Outstanding balance**
		Current year RO	**Carried over RO**	**Total**	**Current Year RO**	**Carried over RO**	**Total**	
		1	**2**	**3=1+2**	**4**	**5**	**6=4+5**	**7=3-6**
33	Other administrative revenue	0,00	21.519,94	21.519,94	0,00	21.519,94	21.519,94	0,00
60	Single market, innovation and digital	8.065,09	0,00	8.065,09	8.065,09	0,00	8.065,09	0,00
66	Other contributions and refunds	56.925.181,70	0,00	56.925.181,70	56.925.181,70	0,00	56.925.181,70	0,00
Total DG DEFIS		**56.933.246,79**	**21.519,94**	**56.954.766,73**	**56.933.246,79**	**21.519,94**	**56.954.766,73**	**0,00**

TABLE 8 : FINANCIAL IMPACT OF EX-ANTE AND EX-POST CONTROLS in for DG DEFIS

EX-ANTE CONTROLS	Irregularity	Total undue payments recovered
NON ELIGIBLE IN COST CLAIMS	4.646.121,58	**4.646.121,58**
CREDIT NOTES	1.586.188,83	**1.586.188,83**
RECOVERY ORDERS ON PRE-FINANCING		
Sub-Total	**6.232.310,41**	**6.232.310,41**

EX-POST CONTROLS	Irregularity	Total undue payments recovered
INCOME LINES IN INVOICES		
RECOVERY ORDERS OTHER THAN ON PRE-FINANCING		
Sub-Total		
GRAND TOTAL (EX-ANTE + EX-POST)	**6.232.310,41**	**6.232.310,41**

TABLE 9: AGEING BALANCE OF RECOVERY ORDERS AT 31/12/2022 for DG DEFIS

	Number at 01/01/2022	Number at 31/12/2022	Evolution	Open Amount (Eur) at 01/01/2022	Open Amount (Eur) at 31/12/2022	Evolution
2021	1		-100,00 %	21.519,94		-100,00 %
	1		**-100,00 %**	**21.519,94**		**-100,00 %**

TABLE 10 :Recovery Order Waivers >= 60 000 € in 2022 for DG DEFIS

Waiver Central Key	Linked RO Central Key	RO Accepted Amount (Eur)	LE Account Group	Commission Decision	Comments

Total DG DEFIS	

Number of RO waivers	

TABLE 11 : Negotiated Procedures in 2022 for DG DEFIS

Negotiated Procedure Legal base	Number of Procedures	Amount (€)
Negotiated procedure with single tender Annex 1 - 11.1	6	400.341.724,00

Total	**6**	**400.341.724,00**

The increase in the number of negotiated procedures is due to the signature of 5 individual contracts with 5 Member States entities for the same single action. Considering that these contracts have the same purpose and have been concluded on the same legal basis, this can be considered as a one-off negotiated procedure. It is also due to the award of a framework contract for Copernicus launchers with Arianespace for € 400.000.000, which will be implemented by ESA in indirect management. It is worth noting that this framework contract does not imply any financial commitment at Commission level and therefore did not appear as such in the standard business objects reports. Consequently, DG DEFIS does not deem any measures necessary as there is no continuous trend to use negotiated procedures.

TABLE 12 : Summary of Procedures in 2022 for DG DEFIS

Internal Procedures > € 60,000

Procedure Legal base	Number of Procedures	Amount (€)
Open procedure (FR 164 (1)(a))	8	20.126.721,00
Negotiated procedure with single tender Annex 1 - 11.1	6	400.341.724,00
Total	**14**	**420.468.445,00**

TABLE 13 : BUILDING CONTRACTS in 2022 for DG DEFIS

Legal Base	Procedure subject	Contract Number	Contractor Name	Contract Subject	Contracted Amount (€)

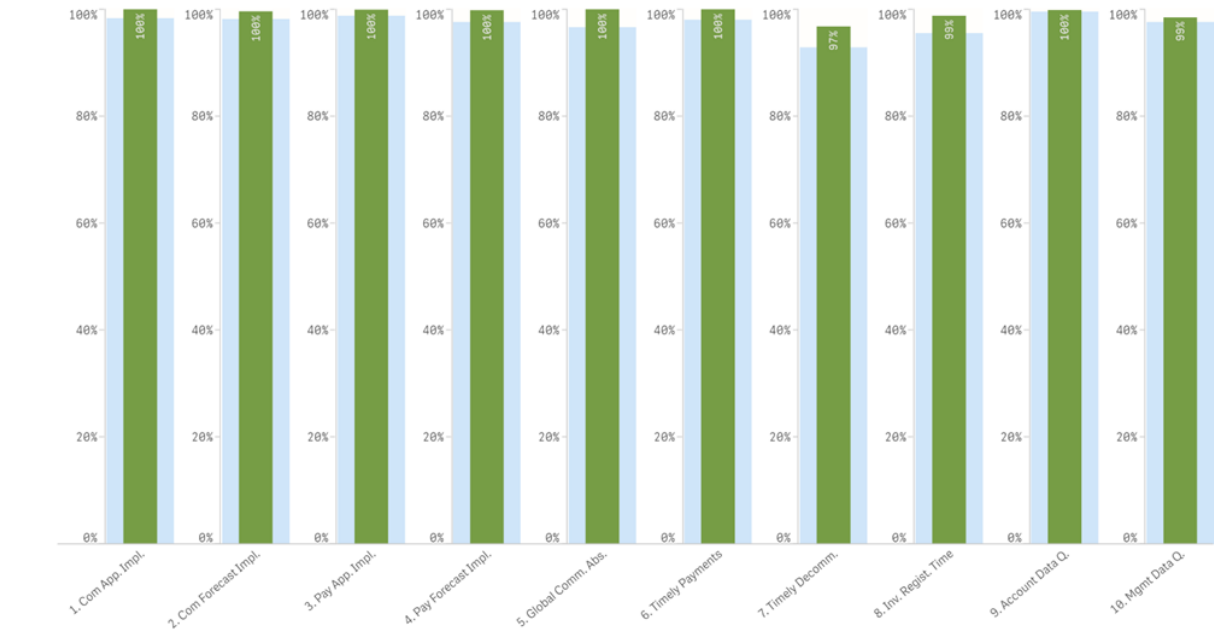
TABLE 14 : CONTRACTS DECLARED SECRET in 2022 for DG DEFIS

Legal Base	LC Date	Contract Number	Contract Subject	Contracted Amount (€)

TABLE 15 : FPA duration exceeds 4 years - DG DEFIS

TABLE 16 : Commitments co-delegation type 3 in 2022 for DG DEFIS

ANNEX 4: Financial scorecard



For each indicator the light blue bar denotes the EC Score.

Indicator	Objective	Comment[31]	DEFIS Score	EC Score
1. Commitment Appropriations Implementation	Ensure efficient use of commitment appropriations expiring at the end of Financial Year	N/A	100%	98%
2. Commitment Forecast Implementation	Ensure the cumulative alignment of the commitment implementation with the commitment forecast in a financial year	N/A	100%	98%
3. Payment Appropriations Implementation	Ensure efficient use of payment appropriations expiring at the end of Financial Year	N/A	100%	99%
4. Payment Forecast Implementation	Ensure the cumulative alignment of the payment implementation with the payment forecast in a financial year	N/A	100%	98%
5. Global Commitment Absorption[32]	Ensure efficient use of already earmarked commitment appropriations (at L1 level)	N/A	100%	97%

[31] An explanation behind the indicator result can be provided, e.g. the comment about the achievement itself, reference to the whole Commission performance (better or worse), reasons behind this achievement. The comment is mandatory for the 'Timely payments' indicator. For the rest of indicators the comment is mandatory only if the score is equal or below the target of 80%.

[32] Due to technical limitation: 1. the indicator does not take into account the Com L1 Consumption between the FDC ILC date and the FA FDI allowed as an exception in the external actions for Com L1 of type GF, i.e. with Financing Agreement, under the FR2018 Article 114.2. 2. it is

6. Timely Payments	Ensure efficient processing of payments within the legal deadlines	N/A	100%	98%
7. Timely Decommitments	Ensure efficient decommitment of outstanding RAL at the end of commitment life cycle	N/A	97%	93%
8. Invoice Registration Time	Monitor the accounting risk stemming from late registration of invoices in the central accounting system ABAC	N/A	99%	95%
9. Accounting Data Quality	Ensure the good data quality of ABAC transactions with the focus on fields having a primary impact on the accounts	N/A	100%	100%
10. Management Data Quality	Ensure the good data quality of ABAC transactions with the focus on fields having a primary impact on the management decisions	N/A	99%	98%

technically not possible to exclude the decommitment of RAL (C8) which is subsequently re-committed for a new purpose. As a result, the actual Indicator score may be slightly higher than the one reported for DGs using the GF commitments.

ANNEX 5: Materiality criteria

Without prejudice of maintaining a reservation for its reputational reasons if applicable.

The present document details the way DEFIS assesses the level of errors in its annual financial statements and the definition of the level of misstatement that is considered as quantitatively material. The overall control objective is to ensure that the residual error rate affecting the relevant expenditure of 2022 remains below 2%.

DEFIS's expenditure is composed of (in order of importance), indirectly managed procurement (space) and grants (H2020, Horizon Europe), directly managed grants (EDIDP, EDF) and other direct spending mostly of an administrative nature. The error rate affecting payments is estimated yearly and per management system, following a methodology that takes into account the risk associated to the type of expenditure (in terms of probability and final financial impact).

DEFIS managed in 2022 financial operations under GNSS (Galileo, Egnos), Copernicus, H2020 and other programmes such as EDF and EDIDP. Considering that the biggest part of DEFIS's yearly expenditure is related to indirectly managed Space Delegation agreements, the following section focusses on this specific management system.

A. Space programmes (GNSS, Copernicus)

The assessment of the effectiveness of the different programmes' control system is based on ex-ante verification and on ex-post audits' results. The effectiveness is expressed in terms of detected and residual error rate, calculated from the best available estimates.

Assurance on the efficient use of delegated fund is obtained though the following elements:

- Before signing a Contribution agreement (Delegation agreement under previous MFF), the Entrusted Entity undergoes a pillar assessment on its internal control environment
- Pre-financing payments are subject to ex-ante operational and financial verification of the quarterly/semestrial and annual implementation reports (respectively QIRs/SIRs and AIRs) provided by entrusted entities; assessing the reliability of budgetary data; keeping an up to date overview of commitments and payments forecasted and executed; ensuring that actual and forecasted expenses remain within the allocated budget; checking actual expenditure versus forecast, and updating cash needs and level of commitments; developing a long-term financial planning covering the full MFF; verifying consistency of figures and challenging costs reported based on information provided; justifying acceptance of costs and clearings, analysing the schedule, its credibility and challenging it, if needed; analysing the risks and the effectiveness of the mitigation actions proposed; analysing the KPIs and ensuring they are in the limits requested by the programme.

- In depth ex-ante controls of accounting data, ex-ante checks of reported costs on assets at year end (review of status of contracts)
- Ex-post audits performed by DG DEFIS on annual declared cost. For ESA and EUSPA, all Annual Implementation Reports under the different programmes are audited every year. The other Entrusted Entities are audited “at least” every two years.

The errors detected during the audit have no impact on the legality and regularity of the amounts paid to the entities, because final annual eligible costs, taking into account audit adjustments, are cleared from the total outstanding pre-financing amount.

The error rates used for the calculation of the amount at risk at payments for each programme are the detected error rates stemming from the finalized ex-post audits performed for the respective year. The detected error rate is calculated as a comparison between the errors and the amount of the audited sample of the AIR. As all errors are corrected through adjustments in the clearing, the detected error rate for Space programmes is equal to the residual error rate.

DG DEFIS considers that the error rates detected by its audits are a reliable indicator for the legality and regularity of the non-audited transactions. Furthermore, based on its monitoring and supervision, DG DEFIS considers that the level of error remains relatively stable over the years. Even if the error rate happens to be very low DG DEFIS decided to use an error rate of 0.5% as a conservative approach.

B. Research framework programme H2020

The assessment of the effectiveness of the different programmes' control system is based mainly, but not exclusively, on ex-post audits' results. The effectiveness is expressed in terms of detected and residual error rate, calculated on a representative sample on a multi-annual basis.

Assessment of the effectiveness of controls

The starting point to determine the effectiveness of the controls in place is the cumulative level of error expressed as the percentage of errors in favour of the EC, detected by ex-post audits, measured with respect to the amounts accepted after ex-ante controls.

However, to take into account the impact of the ex-post controls, this error level is to be adjusted by subtracting:

- Errors detected corrected as a result of the implementation of audit conclusions.
- Errors corrected as a result of the extension of audit results to non-audited contracts with the same beneficiary.

This results in a residual error rate, which is calculated in accordance with the following formula:

$$\mathrm{Re}\,sER\% = \frac{(\mathrm{Re}\,pER\% * (P - A)) - (\mathrm{Re}\,pERsys\% * E)}{P}$$

where:

ResER% residual error rate, expressed as a percentage.

RepER% representative error rate, or error rate detected in the common representative sample, expressed as a percentage. The RepER% is composed of complementary portions reflecting the proportion of negative systematic and non-systematic errors detected. This rate is the same for all implementing entities, without prejudice to possibly individual detected error rates.

RepERsys% portion of the RepER% representing negative systematic errors, (expressed as a percentage). The RepERsys% is the same for all entities and it is calculated from the same set of results as the RepER%

P total requested EC contribution (€) in the auditable population (i.e. all paid financial statements).

A total requested EC contribution (€) as approved by financial officers of all audited financial statements. This will be collected from audit results.

E total non-audited requested EC contribution (€) of all audited beneficiaries.

The Common Representative Sample (CRS) is the starting point for the calculation of the residual error rate. It is representative of the expenditure of each FP as a whole. Nevertheless, the Director-General (or Director for the Executive Agencies) must also take into account other information when considering if the overall residual error rate is a sufficient basis on which to draw a conclusion on assurance (or make a reservation) for specific segment(s) of the Seventh Framework Programme (FP7)/Horizon 2020. This information may include the results of other ex-post audits, ex-ante controls, risk assessments, audit reports from external or internal auditors, etc. All this information may be used in assessing the overall impact of a weakness and considering whether to make a reservation or not.

If the CRS results are not used as the basis for calculating the residual error rate this must be clearly disclosed in the AAR, along with details of why and how the final judgement was made.

Should a calculation of the residual error rate based on a representative sample not be possible for a FP for reasons not involving control deficiencies,[33] the consequences are to be assessed quantitatively by making a best estimate of the likely exposure for the reporting year based on all available information. The relative impact on the Declaration of Assurance would then be considered by analysing the available information on qualitative grounds and considering evidence from other sources and areas. This should be clearly explained in the AAR.

Multi-annual approach

The Commission's central services' guidance relating to the quantitative materiality threshold refers to a percentage of the authorised payments of the reporting year of the ABB expenditure. However, the Guidance on AARs also allows a multi-annual approach, especially for budget areas (e.g. programmes) for which a multi-annual control system is more effective. In such cases, the calculation of errors, corrections and materiality of the residual amount at risk should be done on a "cumulative basis" on the basis of the totals over the entire programme lifecycle.

Because of its multiannual nature, the effectiveness of the Research and Innovation family services' control strategy can only be fully measured and assessed at the final stages in the life of the framework programme, once the ex-post audit strategy has been fully implemented and systematic errors have been detected and corrected.

In addition, basing materiality solely on ABB expenditure for one year may not provide the most appropriate basis for judgements, as ABB expenditure often includes significant levels of pre-financing expenditure (e.g. during the initial years of a new generation of programmes), as well as reimbursements (interim and final payments) based on cost claims that 'clear' those pre-financings. Pre-financing expenditure is very low risk, being paid automatically after the signature of the contract.

Notwithstanding the multiannual span of their control strategy, the Directors-General of the Research DGs (and the Directors of the Executive Agencies implementing Research and Innovation Framework Programmes are required to sign a statement of assurance for each financial reporting year. In order to determine whether to qualify this statement of assurance with a reservation, the effectiveness of the control systems in place needs to be assessed not only for the year of reference but also with a multiannual perspective, to determine whether it is possible to reasonably conclude that the control objectives will be met in the future as foreseen.

In view of the crucial role of ex-post audits defined in the respective common audit strategies, this assessment needs to check in particular whether the scope and results of the ex-post audits carried out until the end of the reporting period are sufficient and adequate to meet the multiannual control strategy goals.

([33]) Such as, for instance, when the number of results from a statistically-representative sample collected at a given point in time is not sufficient to calculate a reliable error rate.

The criteria for making a decision on whether there is material error in the expenditure of the DG or service, and thus, on whether to make a reservation in the AAR, will therefore be principally, though not necessarily exclusively, based on the level of error identified in ex-post audits of cost claims on a multi-annual basis.

Adequacy of the audit scope

The quantity of the (cumulative) audit effort carried out until the end of each year is measured by the actual volume of audits completed. The data is to be shown per year and cumulated, in line with the current AAR presentation of error rates. The multiannual planning and results should be reported in sufficient detail to allow the reader to form an opinion on whether the strategy is on course as foreseen.

The Director-General (or Director for the Executive Agencies) should form a qualitative opinion to determine whether deviations from the multiannual plan are of such significance that they seriously endanger the achievement of the internal control objective. In such case, she or he would be expected to qualify his annual statement of assurance with a reservation.

Revised methodology for the calculation of the error rate for Horizon 2020

European Court of Auditors observations

The European Court of Auditors observed in its 2018 and 2019 Annual Reports that the error rate of Horizon 2020 was understated due to the fact that the "*ex-post audits aim for maximum coverage of the accepted costs, but rarely cover all the costs. The error rate is calculated as a share of all the accepted costs, instead of the amount actually audited. This means that the denominator in the error calculation is higher, so the error rate is understated. In case the errors found are of a systemic nature, the error is extrapolated which partially compensates for the above-mentioned understatement. However, since extrapolation is not performed for non-systemic errors, the overall error rate is nevertheless understated. The understatement of the error rate cannot be quantified. It is, then, impossible to determine whether the impact of this understatement is significant*".

In response to this observation, in 2020 the Commission re-defined its methodology for calculating the Horizon 2020 error rate. In order to quantify any potential understatement mentioned by the Court, the Commission applied a new methodology for all audits closed as from 01 January 2020. The main change in the methodology is that, the denominator used in the error calculation is the sum of costs actually audited and not the sum of all accepted costs.

In this respect, an additional 0.37 % (calculated on 1 304 H2020 audit participations by difference with the previous methodology) has been used to top up the cumulative detected error rate for 2021.

IAS limited review on the 2020 error rate calculation for H2020

The IAS has carried out a limited review on the methodology for calculation of the error rates of Horizon 2020 in year 2020. The findings of this limited review confirmed that there is no weakness in the calculation of the detected error rate and that the impact of these findings on the accuracy of the calculation of the residual error rate is minor. The IAS recommended that:

1. The CIC should:
- 1.1 Calculate the corporate H2020 residual error rate based on the actual level of implementation of audit results and extension of audit findings stemming from data encoded by the Authorising Officers;;
- 1.2 Considering that there is no data on the sampled amounts for audits closed before 2020, estimate the amount actually audited by calculating the ratio of costs actually audited to the total amount of the related accepted cost claims for all the audits closed since 1 January 2020 ('A' parameter in the formula for calculating the residual error rate) and adapt parameter 'E' accordingly;
- 1.3 Formalise the changes in the residual error rate calculation (e.g. in a written CAS procedure).
2. The CAS should:
- 2.1. Change the audit report template to include a line in the table of Annex 1 with the audited amounts (sampled);
- 2.2. Include fields in AUDEX to encode the audited amounts per participation and cost category and any other IT tool used to register the ex post audits' data which feed the Microsoft Access database for the calculation of the representative detected error rate;
- 2.3. Calculate the top-up automatically in the Microsoft Access database. For Horizon Europe, the calculation will be in line with the new methodology and no top up calculation will be required.

The recommendations 1.1 to 2.2 above are fully implemented. The recommendation 2.3 which refers to Horizon Europe and will be completed once the audit campaign for Horizon Europe start. Nevertheless, the new methodology, without the need for a top up calculation, is already being implemented for H2020 audits.

C. Other programmes

For Horizon Europe, the general control objective, following the standard quantitative materiality threshold proposed in the standing instructions for Annual Activity Reports, is to ensure that the cumulative residual error rate, i.e. the level of errors which remain undetected and uncorrected, does not exceed 2%.

For all other programmes implemented by DEFIS (EDIDP i.e.) the general objective is to ensure that the residual error rate, i.e. the level of errors which remain undetected and uncorrected, does not exceed 2% by the end of the programmes' management cycle.

D. De ‘minimis' threshold for financial reservations

“Since 2019[34], a 'de minimis' threshold for financial reservations has been introduced. Quantified AAR reservations related to residual error rates above the 2% materiality threshold, are deemed not substantial for segments representing less than 5% of a DG’s total payments and with a financial impact below EUR 5 million. In such cases, quantified reservations are no longer needed."

([34]) Agreement of the Corporate Management Board of 30/4/2019.

ANNEX 6: Relevant Control System(s) for budget implementation (RCSs)[35]

RCS N°1: Budget entrusted to other entities

This RCS covers: (1) under indirect management, the agreements with **ESA** for the EGNSS, Galileo FOC, EGNOS, Copernicus, SSA, GOVSATCOM, IOD/IOV (2) under indirect management, the agreement with **EUSPA** for the space programme components, horizontal activities, research activities and past agreements for EGNSS (3) under indirect management, Copernicus agreements with **EUMETSAT, ECMWF, EEA, EMSA, FRONTEX, MERCATOR OCEAN, SATCEN** (3) under indirect management, the agreements with **OCCAR** and **EDA** for defence industry programmes; (4) cross sub-delegations to **other Commission services** (such as JRC).

[35] DG DEFIS Control Strategy approved on 16/02/2022, Ares(2022)1145348.

Stage 1 – Establishment (or prolongation) of the mandate to the Entrusted Entity (EE)

Main control objectives: Ensure that the legal framework for the management of the relevant funds is fully compliant and regular (legality & regularity), that is delegated to the appropriate entity (best value for public money, economy, efficiency), without any conflicts of interests (anti-fraud strategy) and gives all the rules necessary for a smooth running by the EE.

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
The Financial Framework Partnership Agreement (FFPA) with EUSPA and ESA and Contributions Agreement (CA) does not clearly set out: - delegated tasks, responsibilities of each involved actor - internal control and reporting requirements to be observed - arrangements for protection of EU financial interests and transparency of operations - right of the European Court of Auditors (ECA) and the European Anti-Fraud Office (OLAF) to comprehensively exert their competences to audit the entrusted funds Specific risks related to industrial procurement to be carried out by ESA	Take the *lessons learned* from prior similar CAs Ex-ante review by the relevant staff within DG DEFIS Consultation of the central EU services (DG BUDG, Legal Service) Hierarchical validation within the authorising directorate Describe modalities of cooperation, supervision and reporting in the FFPA and CA Explicit allocation of supervision responsibility to individual officials (reflected in task assignment or function descriptions) Ex-ante verification of the procurement's procedures carried out by the EE in their own name (for example: EC procurement board with ESA), or on behalf of the EU Scrutiny verification by DG DEFIS of	**Coverage/Frequency**: 100 %/once **Depth**: Checklist includes a list of the requirements of the regulatory provisions to be complied with. <u>Factors</u> would be (i) whether it is an establishment or a prolongation, (ii) whether it involves selecting an entity and (iii) consistency with any other entities entrusted by the same DG or family.	**Costs**: estimation of FTEs involved in the preparation and adoption work **Benefits**: - Total budget amount entrusted to the entity in case of detection of no significant (legal) errors - DG DEFIS reputation intact **Effectiveness**: - Quality of the legal work (Basic Act, Legal and Financial Statement and CA) - no ECA or OLAF observations **Efficiency**: - Average cost of preparation, adoption work done compared with similar cases as benchmark **Economy:** - ratio FTEs/funds entrusted (economic when below 10-15 %)

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
on behalf of DG DEFIS in the complex oligopolistic space market Specific risks related to industrial procurement to be carried out by EUSPA in the complex oligopolistic space market	industrial procurements procedures carried out by EUSPA		

Stage 2 – Ex-ante (re)assessment of the Entrusted Entity's financial and control framework

Main control objectives: Ensuring that the EE is fully prepared to start/continue implementing the delegated funds autonomously as requested by the Financial Regulation (Art. 154.4).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
Before entrusting tasks of budget implementation to the EE, DG DEFIS has not obtained evidence that the financial and control framework deployed by the EE is sufficiently mature The EE's own financial framework differs from the EU FR and the two parallel systems coexist with the risk of the EE's own system being applied to EU funds The EE has not timely informed DG DEFIS about substantial changes made to its systems, rules and procedures that relate to the management of the EU funds entrusted	DG DEFIS internal or independent external ex-ante assessment of the EE ensuring that there is the same level of protection of the financial interests of the Union equivalent to the one that is provided for when the Commission implements the EU budget (Article 62 FR) Hierarchical validation within the authorising directorate Require justification and prior consent for any deviation to financial rules (e.g., Riders or Contract Change Notices) Require timely notification by the EE of any changes to its financial or control systems subsequent to the signature of the CA Statement obtained from another DG which also has a CA with the EE (cross-reliance of pillar assessments and exemptions)	**Coverage/frequency**: - International organisations: prior to the signature, assessment of internal control system of the EE followed if necessary by ad hoc targeted system controls - Agencies: targeted system controls/ad hoc **Depth:** - 100 %	**Costs**: - estimation of FTEs involved in the ex-ante assessment process (including missions) - cost of outsourced independent external "pillar" (re)assessment of the EE's control system(s), if not covered by the EE **Benefits**: - Total budget amount entrusted to the EE if no significant system weaknesses are detected - DG's reputation remains intact **Effectiveness**: - no ECA or IAS observations - n° of recommendations proposed to EE as result of assessment (i.e., deviations from EU FR identified) - quality of ex-ante assessment **Efficiency Indicators**: - Time-To-Implement recommendations (by the EE) - Time-To-(Re)Assess **Economy:** ratio FTEs/funds

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
			entrusted (economic when below 10-15 %)

Stage 3 – Implementation: monitoring, supervision, reporting

Main control objectives: Ensure that the FFPA/CA objectives are achieved and that DG DEFIS is fully and timely informed of any relevant management issues encountered by the EE, in order to possibly mitigate potential financial and/or reputational impacts (legality & regularity, sound financial management, true and fair view reporting, anti-fraud strategy).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- Low quality programme results, delayed programme implementation, non-achievement of policy objectives / desired impact on society. - Due to weak modalities of cooperation, supervision and reporting, DG DEFIS is not fully and timely informed of relevant financial and/or management issues encountered by the EE, and/or does not (timely) react upon notified issues by mitigating them or by making a reservation for them – which may reflect negatively on the DG's governance reputation and quality of accountability reporting. - EE's financial and control systems are not	Detailed reporting modalities included in FFPA/CA (incl. regular programme evaluation). Reinforced monitoring: - increased participation in EE's governance bodies and technical committees - detailed analysis of all reports submitted by the EE; if necessary, request additional ad hoc reports - outsourcing of technical assistance on general programme management and ad hoc topics (e.g., asset management, systems audits) - regular EE audits by DG DEFIS, IAS, ECA and close follow-up of implementation of audit recommendations - management review of the supervision results (e.g., regular DEFIS-ESA meeting at Director-General	**Coverage**: 100 % of the entities are monitored/supervised. **Frequency**: - daily (operational/financial/ technical issues) - monthly (briefings and reports for high level governance meetings) - quarterly (report analysis) - annual (AOXD reports, review of Annual Reports for reservations) <u>In case of</u> operational / financial issues, measures are reinforced. The **depth** depends on the <u>mandate</u> given to the entity, and on the level of DG DEFIS access to the EE's internal control information.	**Costs**: estimation of FTEs involved in monitoring and supervision (including missions). **Benefits**: - Total budget amount entrusted to the EE if no significant (legal, management, accounting, fraud, reporting) errors are detected - DG's reputation remains intact **Effectiveness**: - FFPA/CA objectives achieved on time - cut-off and closure exercise carried out within deadline - relevance, reliability and quality of control data reported back by EE - n° of serious IAS or ECA findings on control failures - n° of regular monitoring actions, n° of issues under reinforced monitoring, budget % value and amount of errors

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
functioning as expected, even though the outcome of the system (re)assessment was satisfactory (e.g., assets not correctly registered in EEs accounts) - EE's procedures are changed during the mandate	level) - monthly EC/ESA/EUSPA directors meetings to tackle specific issues - set up of ad hoc DEFIS - EE Task Forces to tackle problematic issues - if necessary, referral to OLAF - DG DEFIS is informed in due time of changes in order to assess the impact on the implementation of EU funds and agree or not on changes		detected ex-post - Parent DG's AAR assurance on EEs budgets **Efficiency Indicators**: - no amendments to FFPA/CA to extend programme implementation deadline - FFPA/CA renewed - Time-To-Implement audit recommendations **Economy:** - ratio FTEs/funds entrusted (economic when below 10-15 %)

Stage 4 – Commission contribution: payment or suspension/interruption

Main control objectives: Ensure that the Commission fully assesses the management situation at the Entrusted Entity (EE), before either paying out the (next) financial contribution for the operational and/or operating budget of the entity, or deciding to suspend/interrupt the (next) contribution (legality & regularity, sound financial management, anti-fraud strategy).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
The Commission pays out the (next) contribution to the Entrusted Entity: - while not being aware of management issues that may lead to financial and/or reputational damage - despite being aware of such issues - with incorrect calculation of the cash needs of the Entrusted Entity with no implementation of the audit results by the Entrusted Entity	- Require EE to report back on management issues as soon as possible - Ex-ante operational and financial verifications leading to correction of errors and restatement of corrected contribution request - Management review of supervision results - Hierarchical validation of contribution payment and recovery of non-used funds If necessary, suspension or interruption of payments	**Coverage**: 100 % of the contribution payments. **Frequency**: as per transfer agreement or transfer request The **depth** depends on the mandate of the (type of) entity, inter alia whether DG DEFIS has full access to the entity's internal control information.	**Costs**: estimation of FTEs involved in the ex-ante verifications **Benefits**: - value of errors detected by ex-ante controls - Total budget amount entrusted to the entity if no significant (legal, management, accounting, fraud, reporting) errors are detected - DG's reputation remains intact **Effectiveness**: - amount of unused operating budget recovered (if any) - budget amount of the suspended/interrupted payments (if any). **Efficiency Indicators**: - Time-To-Pay /Recover **Economy:** - Ratio FTEs/funds entrusted (economic when below 10-15 %)

Stage 5 – Audit and evaluation

Main control objectives: Ensuring that assurance building information on the EE's activities is being provided through independent sources as well, which may confirm or contradict the management reporting received from the Entrusted Entity itself.

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- The Commission has insufficient information from independent sources on the EE's management achievements, which prevents drawing conclusions on the assurance for the budget entrusted to the Entity – which may reflect negatively on the Commission's governance reputation and quality of accountability reporting - Decentralised agencies do not fully cooperate with the Discharge authorities and do not provide, as appropriate, any necessary additional information - The entrusted AOXD's control system is subject to AAR reservations and/or ECA criticism - EE refuses access to facilities/information on the basis of classification of the information	- FFPA/CA to specify independent audit function and cooperation with IAS and ECA - DG DEFIS own on-the-spot ex-post audits of the EE and/or its beneficiaries - potential escalation of any major governance-related issues - Interim evaluations by independent experts of achievement of policy objectives - if necessary, refer to OLAF - auditors in possession of a personal security clearance	**Coverage**: All delegation agreements are checked through samples. Regarding the subsidy paid to EUSPA, the budget executed on behalf of DG DEFIS, is checked by the European Court of Auditors. DG DEFIS does not perform ex-post audits on this subsidy of this agency. The AOXDs' systems are presumed to be up to Commission standards. **Frequency**: once a year or every second year, depending on the Entrusted Entity The **depth** depends on the <u>mandate</u> of the (type of) entity, inter alia whether the Commission has full access to the entity's internal control information.	**Costs**: - estimation of FTEs involved in the coordination and execution of the own audits - Ex-post audit mission costs - Cost of outsourced audits **Benefits**: - Assurance of the AOD that the population audited is clean of error - % rate and value of errors detected by own audits (and subsequently corrected) **Effectiveness**: - unqualified opinion by the EE's independent external auditor on the EE's annual financial statements - detected error rate of own ex-post audits of EE below materiality threshold - n° of own audits - n° and amount of errors detected by own audits **Efficiency**: - value of total payments audited - Number of audits launched in the year versus annual target

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
			- Number of audits closed in the year versus annual target **Economy:** - ratio: annual cost of own audits / amount of all errors detected - average cost per audit

RCS N°2: Assets

This RCS covers: the physical assets of the EGNSS and Copernicus space programmes

Stage 1 – Recognition: establishment of the Commission's rights on assets in the underlying agreements

Main control objectives: Negotiation of contractual terms. Ensure that the legal framework (agreements with entrusted entities) for the management of the EU assets is fully compliant and regular (legality & regularity) with an appropriate set-up of requirements related to the safeguarding of assets, inventory management and accounting information (true and fair view).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
FFPA/Contribution Agreement does not clearly set out: - delegated tasks - the requirements related to the ownership, safeguarding and management of EU property - internal control and reporting requirements - arrangements for protection of EU financial interests and transparency of financial operations - right of the European Court of Auditors (ECA) and the European Anti-Fraud Office (OLAF) to comprehensively exert their competences to audit the entrusted funds	Investment of adequate time and effort in drafting the new Contribution Agreement: - Inter-service consultation of relevant Commission services - Detailed and unambiguous modalities of cooperation, supervision and reporting - Stipulations with regard to transfer of ownership and the detailed asset management and reporting requirements	**Coverage/Frequency** 100 %/once **Depth**: In-depth control, full investment of DG DEFIS operational, financial and legal units	**Costs**: estimation of FTEs involved in the preparation and adoption work **Benefits**: - Proper safeguarding of the EU property - DG DEFIS reputation intact - Cost-efficient implementation of the Delegation Agreement **Effectiveness**: - Quality of the legal work (Basic Act, Legal and Financial Statement and DA) - Timely receipt of adequate reporting in line with requirements Delegation Agreements - no ECA, IAS or OLAF criticism **Efficiency**: - Time and average cost of preparation, adoption work done compared with similar cases as benchmark **Economy:** - ratio FTEs/funds entrusted (economic when below 2 %)

Stage 2 – Protection: recording, ensuring correct asset valuation

Main control objectives: Ensuring reliable and accurate asset valuation and reporting (true and fair view).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
The implementation of the Agreements entail weaknesses, which lead to the Commission's legal rights in terms of assets ownerships not being duly protected and/or registered and/or reliably reported Non respect of EU accounting rules regarding assets and inventories Inaccurate valuation of assets	Clear programme specific accounting guidelines, inspection, depreciation and de-commissioning rules Formal agreement of Accounting Officer asked for accounting decisions with a material impact Organisation of asset workshops with the entrusted entities Regular meetings of the asset working group with members from the accounting team, DG for Budget and operational units In depth ex-ante controls of accounting data, including sample-wise ex-ante checks of underlying cost and regular checks of inventories	**Coverage/Frequency:** Full coverage/yearly **Depth**: In-depth control, full investment of DG DEFIS accounting team in co-operation with operational units	**Costs:** estimation of cost of staff involved. Cost of the contracted services (if applicable) **Benefits:** The (average annual) total value of the significant errors detected and thus prevented in terms of the Commission's rights **Effectiveness:** Number of material internal and external audit findings about incorrect valuation of assets The valuation of assets within the deadlines imposed by the Directorate-general for Budget **Efficiency**: Time spent on controls related to the asset value **Economy:** Cost of valuation and accounting of the Commission's assets and evolution over time

Stage 3 – Overall monitoring of proper safeguarding of assets

Main control objectives: Ensuring that the Commission's property is safeguarded adequately.

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
Lack of complete and reliable assets register Lack of safeguarding of assets (for example assets lost, damaged or disposed without prior permission of the EU)	Review the controls strategy with an emphasis on documenting the existing control Monitoring reports provided by the relevant asset manager/inventory Physical inspection of assets under EU ownership Defines formal procedure for disposal of assets Other monitoring measures adequate to the programme (i.e., monitoring of asset performance, signal provision)	Performance of physical inspections on the basis of the multi-annual assets verification programme on a materiality based approach. Annual review of reports provided by the entrusted entities	**Costs:** estimation of cost of staff & missions involved. **Benefits:** assurance on the existence and safeguarding of the total value of EU assets Budget value of items lost detected **Effectiveness:** Value of assets inspected per three years as % of net asset (equipment) value Number of follow-up actions **Efficiency:** Time spent and cost of missions related to the value of assets inspected **Economy:** Cost of inspections of the EU assets and evolution over time

Stage 4 - Ex-Post controls: supervision monitoring, reviews, audits – plus corrections

Main control objectives: Measuring the effectiveness of ex-ante controls; detect and correct any error with regard to the underlying cost remaining undetected after the implementation of ex-ante controls. Ensuring that the appropriate corrections are being made.

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
The ex-ante controls fail to prevent, detect and correct errors in the valuation of the assets	Ex-post audits of cost reported by the entrusted entities that form the basis for the EU asset valuation	Coverage ex post audits: • Representative sample: random or MUS sample sufficiently representative to draw valid management conclusions • Risk-based sample, determined in accordance with the selected risk criteria, aimed to maximise error correction (either higher amounts or expected error rate).	**Costs:** estimation of cost of staff involved in the supervision and audit strategy **Benefits:** budget value of the errors, detected by the auditors, which have actually been corrected. **Effectiveness:** Representative error rate below 2 %. **Efficiency:** total (average) annual cost of audits compared with benefits (ratio). **Economy:** Cost of ex-post audits of the underlying cost of asset valuation and evolution over time

RCS N° 3: Procurement

This RCS covers: DG DEFIS own procurement under direct management, which is mostly for studies and technical assistance:

Stage 1 – Decision to launch a procurement procedure (call for tender)

A – Planning

Main control objectives: Effectiveness, efficiency and economy. Compliance (legality and regularity).

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
- The procurement needs are not clearly defined or justified from an economic or operational point of view - Discontinuation of the services provided due to poor/late planning and organisation of the procurement process - Lack of expert knowledge and experience in the highly regulated field of procurement which may lead to the wrong choice of procedure/threshold s and the splitting of purchases - Conflict of interests	- Publication of intended procurements - Detailed financial procedures - In-house technical trainings on procurement management - Regular information on ethics, integrity and fraud awareness to all staff involved in the procurement process	- 100 % of forecast procurements are included in the DG DEFIS Monitoring Tool.	**Costs**: - estimation of FTEs involved and the related contract values (if external expertise is used) **Quantified Benefits**: - Amount of rejection of unjustified purchases **Non-Quantified Benefits:** - Avoidance of litigation - DG DEFIS reputation intact **Effectiveness**: - n° of ECA observations and % error rate on choice of procurement procedure - n° of successful legal on errors in the procurement procedures

B - Needs assessment & definition

Main control objectives: Effectiveness, efficiency and economy. Compliance (legality and regularity).

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
- Risk of not obtaining value for money due to lack of market analysis and/or poor definition of selection criteria - Risk of unequal treatment resulting in litigation, due to selection criteria favouring one contractor - Risk of not receiving the best offers due to the poor definition of the tender specifications (disproportion between contract value and selection/award criteria, or specifications too vague) - Risk of non-compliance with legality and regularity and criticism on choice of procedure due to limited competition and high proportion of negotiated procedures in the very technical, complex and oligopolistic space market	- Encourage use of open procedures, even in relatively closed markets - Technical specifications are prepared and validated by at least 2 staff members, and approved by the responsible operational HoU/ Director before call launch Verification and validation of tender documents by a specialised team for Public Procurement and Grants Management in the Financial Resources and Internal Control unit before call launch	- 100 % of the specifications are verified. Depth may be determined by the amount and/or the impact on the objectives of the DG if it goes wrong - 100 % of the tenders above a financial threshold (e.g., > EUR 15000) are reviewed. Depth risk-based, depending on sensitivity	**Costs**: - estimation of FTEs involved and the related contract values (if external expertise is used) **Quantified Benefits**: - Value of contracts for which the approval and supervisory control detected material error (negative opinion issued by the DG DEFIS legal team). **Non quantified Benefits**: - Limit the risk of litigation - Limit the risk of cancellation of a tender **Effectiveness:** - N° of negative Public Procurement and Grants Management opinions and n° of favourable opinions with reserves. - N° of 'open' procedures or procedures where only one or no offers were received

C – Evaluation & Award

Main control objectives: Effectiveness, efficiency and economy. Compliance (legality and regularity). Fraud prevention and detection.

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
- The most economically advantageous offer is not selected due to a biased, inaccurate or 'unfair' evaluation process - Over-consumption of resources (human and financial) due to errors or mismanagement leading to award decisions being contested (resulting in Court and Ombudsman cases) Damage to the DG's reputation if fraud or criminal behaviour is discovered (conflict of interest)	- All 'call for tender' evaluations involve the use of opinions of more than one qualified official. The evaluation process is more regulated and formalised as the contract value increases. - Risk based approach: higher risk contracts have more in-depth checks Review of and opinion on evaluation and award documents and process by a specialised legal team before contract award	- Formal evaluation process: Opening and Evaluation committees for all tenders > EUR 140 000[36] including signature of declarations of absence of conflict of interests by the committee members - <u>Risk based approach</u>: 1) second review of evaluation and award documents and process (open and restricted procedure) by the Deputy Director General and Director General for procurements > EUR 10 million 2) validation of negotiated procedures > EUR 140 000 by the Director-General before call launch - 100 % of the offers of the call for tenders are evaluated by more than one qualified official. - 100 % of evaluations are	**Costs**: - estimation of FTEs involved and the related contract values (if external expertise is used) **Quantified Benefits**: - Difference between the most onerous offer and the selected one - N° or value of contracts subject to complaints / irregularities - N° of procurements successfully challenged during standstill period **Non quantified Benefits**: - Compliance with FR - Best value for money **Effectiveness:** - n° of ECA observations and % error rate concerning evaluation & award stage - n° of negative Public Procurement and Grants Management opinions - n° of successful complaints or Ombudsman or Court cases resulting from non-compliant procurement process **Efficiency:**

[36] Thresholds for procurement applicable for procurement procedures launched as of 1st January 2022. EUR 140 000 for procurement of supplies and services, EUR 5 350 000 for the procurement of works and concessions

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
		checked. Depth: required documents provided are consistent	- Time-To-Contract Contract value/cost of FTEs involved in control of contracts

Stage 2 – Contract Management and Financial transactions

Main control objectives: Ensuring that the implementation of the contract is in compliance with the signed contract

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
- Bad or non-execution by the contractor, leading to serious problems if contractual deliveries are critical and no short-term alternatives are available (risk of over-dependency on certain contractors) - The products/services foreseen are not, totally or partially, provided in accordance with the technical description and requirements foreseen in the contract and/or the amounts paid exceed that due in accordance with the applicable contractual and regulatory provisions - Risk of bad execution due to undetected errors on uncorrected imprecisions in offers or tendering specifications - Business discontinues, because contractor fails to deliver - Plagiarism (studies, reports) - Fraud	- Checks on financial capacity and viability of contractors prior to awarding the contract - Close monitoring of contracts, with possible on-site verifications, particularly of high value contracts resulting from negotiated procedures - Checks on both operational and financial issues carried out at appropriate level using the most qualified staff. As defined in the in accordance with the financial circuits - Possibility to run a plagiarism check of reports submitted by contractor Management of sensitive functions	- 100 % of the contracts are controlled, including only value-adding checks - In case of reimbursement costs, in-depth ex-ante verification - High risk operations identified by risk criteria For high-risk operations, reinforced monitoring of the respect of the timely achievement of the contract's milestones by the contractor	**Costs**: - estimation of FTEs involved **Quantified Benefits**: - Amount of irregularities, errors and overpayments prevented by the controls **Non quantified Benefits**: - DG reputation intact **Effectiveness:** - n° of ECA observations and % error rate relating to contract management /payment stage - n° of court cases resulting from contract execution problems - % Budget execution rate – total amount committed/paid versus total budget envelope - % of contracts implemented - n° of open critical and/or very important audit recommendations **Efficiency:** - Time-To-Pay - Late interest payment and damages paid (by the Commission) - Coverage of 1st and 2nd level ex-ante controls **Economy:** - Average n° of contracts per procurement control FTE - cost of control per

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
			running contract % cost over annual amount disbursed

Stage 3 – Supervisory measures

Main control objectives: Ensuring that any weakness in the procedures (tender and financial transactions) is detected and corrected

Main risks	Mitigating controls	Coverage frequency and depth	Cost-Effectiveness indicators (three E's)
- An error or non-compliance with regulatory and contractual provisions, including technical specifications, or a fraud is not prevented, detected or corrected by ex-ante control, prior to payment	- Supervisory desk review of procurement and financial transactions - Ex-post publication of contracts awarded (and subsequent publication in the EU Financial Transparency System) - Regular review of exceptions or non-compliance events reported - Regular review of the procurement process (self-assessment by DG legal team - System and transaction audits by IAS, ECA) and subsequent monitoring of implementation of recommendations for improvement - indicators on procurement are regularly reported	- 100 % Depth: review any significant problem that occurred - legal team examines procurement procedures - 100 % of the sample at least once a year to determine any errors or systemic problems or weaknesses in the procedures (procurement and financial transactions)	**Costs**: - estimation of FTEs involved in the controls **Non-Quantified Benefits**: - Systematic weaknesses corrected - Deterrent effect **Effectiveness:** - Amounts associated with errors detected (related to fraud, irregularities and error) and in % over total checked. - N° system improvements made **Efficiency:** - Average time-to-contract **Economy:** - Proportion of overall cost of control over total expenditure (payments authorised) - Costs of the ex-post controls and supervisory measures with respect to the 'benefits'.

RCS N° 4: Grants

This RCS covers: DG DEFIS grants under direct management (action and operating grants)

Stage 1 – Programming, evaluation and selection of proposals

A - Preparation, adoption and publication of the Annual Work Programme (AWP) and Calls for proposals (Calls)

Main control objectives: Ensuring that the Commission receives and selects the proposals that contribute the most towards the achievement of the policy or programme objectives (effectiveness); Compliance (legality & regularity); Prevention of fraud (anti-fraud strategy)

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- Work Programmes and subsequent calls do not adequately reflect the policy objectives, priorities are incoherent and/or the essential eligibility, selection and award criteria are not adequate to ensure the evaluation of the proposals - Work Programmes are inconsistent within the other family DGs and with the 7 year framework - Work Programmes overlap with other programmes (by other DGs, e.g. Structural Funds) and could lead to double-funding - Calls are tailored to the advantage of certain candidates due to undue influences from interest groups - Calls are not adequately	- Hierarchical validation within the authorising department - Inter-service consultation, including all relevant DGs - Adoption by the Commission - Use corporate templates and corporate tools to prepare and publish the call documents <u>Recommended</u>: - Centralised checklist-based verifications - Explicit allocation of responsibility to individual officials (reflected in task assignment or function descriptions) - Prepare guiding documents and organise infodays and other dissemination events	If risk materialises, all grants awarded during the year under this WP or call would be irregular. Possible impact: 100 % of budget involved and significant reputational consequences. **Coverage**: 100 % **Depth**: All Work Programmes are thoroughly reviewed at all levels, including for operational and legal aspects.	**Costs**: - Estimation of cost of staff involved in the preparation and validation of the Work Programmes and calls. Cost of contracted services, if any. **Benefits**: - Only qualitative benefits. A good Work Programme and well publicised calls should generate a large number of good quality projects, from which the most excellent can be chosen. There will therefore be real competition for funds. - The (average annual) total budgetary amount of the Work Programmes or calls with significant errors detected and corrected. **Effectiveness**: - % of n° of calls successfully concluded / number of calls planned in Management

published and do not reach all target groups - Call documents are incomplete, contain contradicting information or are unclear			Plan/Work Programme - % budget execution rate grant commitments **Economy:** - average n° and value of running grants managed per control FTE - % cost of control for all stages over annual amount disbursed in grants average cost of control per grant

B – Selecting and awarding: Evaluation, ranking and selection of proposals

Main control objectives: Ensuring that the most promising projects for meeting the policy objectives are among the proposals selected (effectiveness); Compliance (legality & regularity); Prevention of fraud (anti-fraud strategy).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- Evaluation, ranking and selection of proposals not carried out in accordance with the established procedures, policy objectives and priorities - Difficulties to recruit adequate independent experts affecting the quality and timing of the evaluations - Eligibility, selection and award criteria too ambiguous or otherwise inadequate to ensure that grants are awarded to the actions which maximise the overall effectiveness of the EU programme - Unauthorised persons may have access to the electronic system for the management of the calls - Unequal treatment of applicants: inappropriate contacts and/or conflict of interests with certain applicants during the procedure	- Selection and appointment of expert evaluators - Outreach actions to achieve a larger pool of available experts - Assessment of evaluation procedure by independent experts - Validation by the AO of ranked list of proposals. In addition, if applicable: opinion of advisory bodies; comitology; inter-service consultation, adoption by the Commission; publication	- 100 % screening of expert evaluators for technical expertise and independence (e.g., conflicts of interests, nationality bias, ex-employer bias, collusion) - 100 % of admissible proposals are evaluated by independent experts - 100 % of contested decisions are analysed by redress committee	**Costs**: - Estimation of cost of staff involved in the evaluation and selection of proposals - Cost of the appointment of experts and of the logistics of the evaluation **Benefits**: - 'quality allocation' assurance of the whole committed budget (as it will have been checked ex-ante and is considered reasonable in the interests of the programme) Qualitative benefits: - Expert evaluators from outside the Commission bring independence, state of the art knowledge in the field and a range of different opinions. This will have an impact on the whole project cycle : better planned, better executed projects **Effectiveness**: - % of proposals evaluated within the year/proposals received - % of n° of (successful) redress challenges / total n° of proposals received - Ratio of proposals

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- Monopoly of certain bodies insufficiently justified			received to proposals selected ("oversubscription" rate) - No successful litigation cases **Efficiency**: - Average Time-To-Publication of selection results **Economy:** - Average evaluation cost per proposal (external experts paid only) - % cost of control over annual amount disbursed in grants

Stage 2 - Contracting

Main control objectives: Ensuring that the most promising projects for meeting the policy objectives are among the proposals contracted; Ensuring that the actions and funds allocation is optimal (Sound Financial Management: best value for public money; effectiveness, economy, efficiency); Compliance (legality & regularity); Prevention of fraud (anti-fraud strategy)

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- After evaluation, the description of the action in the grant agreement remains unclear or still includes tasks which do not contribute to the achievement of the programme objectives - Inconsistencies exist between the grant agreement and its annexes - Procedures do not comply with regulatory framework - The consortium proposes changes that put into question the result of the evaluation - Delays in the provision of the necessary input to finish the GAP, putting at risk the signature of the contract before year N+1 - The beneficiary : ▪ has overestimated the costs necessary to carry out the action ▪ has made false declarations ▪ lacks operational	- Systematic checks on operational and legal aspects performed before signature of the grant agreement - Project Officers implement evaluators' recommendations in discussion with selected applicants. Hierarchical validation of proposed adjustments. - Validation of beneficiaries (operational and financial viability) - Planning of (mid-term and final) evaluations. - Signature of the grant agreement by the AO. - In-depth financial verification and taking appropriate measures for high risk beneficiaries - Preparation of a reserve list of projects	**Coverage:** - 100 % of the selected proposals and beneficiaries are scrutinised - 100 % of draft grant agreements **Depth** may be differentiated; determined after considering the type or nature of the beneficiary (e.g. SMEs, joint-ventures) and/or of the modalities (e.g. substantial subcontracting) and/or the total value of the grant	**Costs**: - Estimation of cost of staff involved in the contracting process **Efficiency**: - Average Time-To-Grant **Economy:** - % cost of control for all stages over annual amount disbursed in grants

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
and/or financial capacity to carry out the action ▪ is awarded several grants for a single action (double-funding by different DGs or other donors) ▪ falls into an exclusion situation			

Stage 3 - Monitoring the execution: Project management - operational, financial and reporting aspects

Main control objectives: ensuring that the operational results (deliverables) from the projects are of good value and meet the objectives and conditions (effectiveness & efficiency); ensuring that the related financial operations comply with regulatory and contractual provisions (legality & regularity); prevention of fraud (anti-fraud strategy); ensuring appropriate accounting of the operations (reliability of reporting, safeguarding of assets and information).

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- The actions foreseen are not, totally or partially, carried out in accordance with the technical description and requirements foreseen in the grant agreement and/or the amounts paid exceed those due according to the applicable contractual and regulatory provisions. - Eligibility conditions of the beneficiary may change during the implementation (e.g. SME bought by a larger company or a company becoming controlled by a third state) - Reimbursement of ineligible costs by DG DEFIS (e.g. due to overinflated timesheets, subcontracting of core activities or without prior tendering procedure)	- Kick-off meetings and "launch events" involving the beneficiaries in order to avoid project management and reporting errors - Explain and clarify at front rules on eligibility criteria (most current cases) - Effective external communication about guidance to the beneficiaries - Operational and financial checks in accordance with the financial circuits. - Operation authorisation by the AO - For riskier operations more in-depth ex-ante controls. Scientific reviews if necessary. - Agreement with Member States on security framework principles and role of Project Manager - When needed: application of	- 100 % of the projects are controlled, including only value-adding checks - Riskier operations subject to more in-depth controls - The depth depends on risk criteria. However, as a deliberate policy to reduce administrative burden, and to ensure a good balance between trust and control, the level of control at this stage is reduced to a minimum. - High risk operations identified by risk criteria. Red flags: suspicions raised by staff, delayed interim deliverables, suspicion of plagiarism, unstable consortium, requesting many amendments, EDES or anti-fraud flagging, etc. - Audit certificates	**Costs**: - estimation of cost of staff involved in the actual management of running projects **Benefits**: - part of budget value of the costs claimed by the beneficiary, but rejected by staff - Reductions in error rates identified by audit certificates - Budget value of penalties and liquidated damages. - Benefits due to operational review of projects and consequent corrective actions imposed on projects **Effectiveness**: - % and value of reductions made to EC contribution paid out through the ex-ante desk checks / total value of cost claims desk-checked - % of payments suspended - n° of cost claims desk-checked **Efficiency**: - % and value of reductions made to EU

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- Several authorising officers implement the same programme and do not treat the beneficiaries equally - Insufficient operational performance monitoring of beneficiaries by project officers - Limitations in the security framework do not allow project officers to access the necessary documents to confirm the correct implementation of the action and to provide the "certified correct"	suspension/interruption of payments, penalties or liquidated damages, earmark projects for risk-based ex-post audit, refer grant/beneficiary to OLAF Desk check (ex-ante) by the financial officers of the cost claims provided by the beneficiaries.	required for any beneficiary claiming significant EU contribution.	contribution paid through ex-ante desk checks/total value of cost claims checked - Average n° & value of projects managed 'per' staff FTE - Average Time-To-Pay - Average payment suspension time (days) **Economy:** - % cost of ex-ante control (cost/total amount of grant payments) - Average project management cost (staff FTE * standard staff cost) per running project

Stage 4 - Ex-Post controls: supervision monitoring, reviews, audits – plus corrections

Main control objectives: Measuring the effectiveness of ex-ante controls; detect and correct any error with regard to the underlying cost remaining undetected after the implementation of ex-ante controls. Ensuring that the appropriate corrections are being made.

Main risks	Mitigating controls	Coverage, frequency and depth	Cost-Effectiveness indicators (three E's)
- The ex-ante controls fail to prevent, detect and correct errors in the implementation of the grants - Beneficiary refuses access to facilities/information on the basis of classification of the information	- Ex-post audits of cost reported by the beneficiaries - Auditors in possession of a personal security clearance	Coverage ex post audits: • Representative sample: random or MUS sample sufficiently representative to draw valid management conclusions • Risk-based sample, determined in accordance with the selected risk criteria, aimed to maximise error correction (either higher amounts or expected error rate). **Frequency**: audit plan to be established in 2022 for 2023	**Costs:** estimation of cost of staff and/or external auditors involved in the supervision and audit strategy **Benefits:** budget value of the errors, detected by the auditors, which have actually been corrected. **Effectiveness:** Representative error rate below 2 %. **Efficiency:** total (average) annual cost of audits compared with benefits (ratio). **Economy:** Cost of ex-post audits of the underlying cost of the grant

ANNEX 7: Specific annexes related to "financial management"

1. Effectiveness: control results

A. International Organisations

European Space Agency (ESA)

In 2022, the second biggest part of DG DEFIS expenditure was delegated to the European Space Agency (ESA) for the implementation of the GNSS (Galileo and EGNOS), the Copernicus, Govsatcom and SSA components of the Space Programme, as well as the Horizon Europe programme, which is totalling 753.2 million EUR (28,66 % of the total DG DEFIS 2022 payments).

The assurance on the effectiveness of the internal control systems with regard to the legality and regularity of the costs reported is built on:

- ***ESA's control results and/or assurance:***
  - Opinion of the external auditor – unqualified audit opinion
  - Statement of Internal Control of the Director-General - clean
  - Reporting quality control at ESA - considered sufficiently reliable to draw conclusions on assurance.
- Authorising Officers by Delegation's own control results on the ESA's operations:
  - Results of the audits of the 2021 Annual Implementation Reports and the implementation of corrections – no error was found on the costs claimed (see further details in Annex 6).

The DG DEFIS ex-post audit team continued to audit all the Annual Implementation Reports (AIRs) and Annual financial reports (AFRs) submitted by ESA in 2022. The results of the previous financial audits have been implemented. These corrections have been made at the time of the annual clearing of pre-financing payments to ESA. The overall residual error rate is below the 2% materiality threshold.

During the monitoring, errors detected ex-ante in the AIRs and the AFRs, have no impact on the legality and regularity of the amounts paid to ESA, because the amounts paid depend both on costs declared and on cash-flows forecasts. There still remains the possibility to rectify them via future clearings of the pre-financings until the end of the delegated programmes.

In the framework of the regular working arrangement and top-level meetings between DG DEFIS and ESA, the Directorate-General closely monitors ESA's progress with the implementation of the programmes and the related reporting. Ensuring business continuity is a priority and in case of service incidents[37], intense efforts are deployed and measures put in place to prevent such critical situations in the future.

The Relevant Control System Template (RCS) for indirect management in Annex 6 demonstrates how the control system put in place in the Directorate-General, addresses the risks related to this type of expenditure.

Result indicators: Indicators of annual error – ESA/GNSS GALILEO

GALILEO FOC & FOC-DC						
Annual Implementation Report of Financial Year	Reported by ESA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2011	342 192 607	340 360 802	1 831 805	0,54 %	1 831 805	0
2012	398 992 495	397 591 998	1 400 497	0,35 %	1 400 497	0
2013	365 152 925	365 065 529	87 396	0,02 %	87 396	0
2014	462 861 925	462 861 925	0	0,00 %	0	0
2015	712 858 582	710 752 128	2 106 454	1.32%	2 106 454	0
2016	389 193 535	389 193 535	0	0,00 %	0	0
2017	404 303 175	404 303 175	0	0,00 %	0	0
2018	287 968 592	287 968 592	0	0,00 %	0	0
2019	188491212	188491212	0	0,00 %	0	0
2020	151 279 719	151 279 719	0	0,00 %	0	0
2021	523 241 364,57	523 241 364,57	0	0,00 %	0	0

Result indicators: Indicators of annual error – ESA/H 2020 /Horizon Europe

H 2020						
Annual Implementation Report of Financial Year	Reported by ESA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2019	34 845 680	34 845 680	0	0 %	0	0
2020	No audit took place					
2021	17.350.540	17.350.540	0	0 %	0	0

(37) For example, the incident occurred in July 2019 when the provision of Galileo Initial Services was impacted by a technical incident in the Galileo ground infrastructure, cfr. supra point 1.1 p.11.

Result indicators: Indicators of error – ESA/COPERNICUS

DG DEFIS ex-post controls covered all Annual Financial Reports (AFR) submitted by ESA. The audit of the 2021 Annual Implementation Report did not reveal any errors. Regular Audits and corresponding corrections ensure that, on a multi-annual basis, the total amount paid under the Delegation Agreement will be compliant with the eligibility rules and will not exceed the limits defined in the Delegation Agreement.

Annual Implementation Report of Financial Year	Reported by ESA (EC accepted costs)	Commission Audit report (eligible costs)	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2012	104 124 840	102 058 630	2 066 210	1,98 %	2 066 210	0
2013	78 518 254	78 524 613	-6 359	0,00 %	-6 359	0
2014	136 135 061	136 133 236	1 825	0,00%	1 825	0
2015	183 930 223	183 930 223	0	0,00%	0	0
2016	651 537 638	651 527 254	3 431	0,00%	3 431	0
2017	556 348 956	556 348 956	0	0,00%	0	0
2018	478 467 614	478 467 614	0	0,00%	0	0
2019	429 300 236	429 300 236	0	0,00%	0	0
2020	339 037 555	339 037 555	0	0,00%	0	0
2021	361 092 666	361 092 666	0	0,00%	0	0

Other international organisations, e.g. MERCATOR, ECMWF, EUMETSAT and OCCAR

The elements that provide assurance on the effectiveness of the internal control system with regard to the legality and regularity are mostly the same as explained under ESA.

In 2022, an audit on the 2021 Annual Implementation Report of **EUMETSAT,ECMWF** was carried out by DG DEFIS. No errors were revealed. No audit was performed on **MERCATOR**, nor **OCCAR** (first pre-financing paid in 2020).

These entrusted entities are normally audited every two years, according to the audit guidelines established at DG DEFIS, unless specific circumstances justifying a change, are identified.

Result indicators: Indicators of error – EUMETSAT/COPERNICUS

Annual Implementation Report of Financial Year	Reported by Eumetsat	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2013	339 061	341 708	- 2 647	0	- 2 647	0
2014	5 788 694	5 815 842	- 27 148	0	- 27 148	0
2015	20 494 837	Not audited				
2016	27 779 921	27 085 987	693 933	2.5%	693 933	0
2017	Not audited					
2018	43 353 898	44 353 898	3 889.81	0,00%	3 889.81	0
2019	Not audited due to lack of staff					
2020	Not audited					
2021	64 801 980	64 801 980	0	0,00%	0	0

Result indicators: Indicators of error – MERCATOR/COPERNICUS

Annual Implementation Report of Financial Year	Reported by MERCATOR	Commission Audit report	Adjustment	Detected error rate
2014	181.128	180.341	-787	0.43%
2015	11.323.190	11.310.953	-12.237	0.11%
2016	19.328.239	19.319.822	-8.417	0.11%
2017	Not audited			
2018	24.587.119	24.579.863	-7.256	0
2019	Not audited			
2020	25.782.438	25.777.751	-4.687	99,97%
2021	Not audited			

Result indicators: Indicators of error – ECMWF/COPERNICUS

Annual Implementation Report of Financial Year	Reported by ECMWF	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2017	41 836 912	41 836 912	0	0.%	0	0
2018	Not audited					
2019	54 714 886	54 714 886	0	0.%	0	0
2020	Not audited					
2021	48 536 509	48 536 509	0	0.%	0	0

B. EU Agencies

Decentralised Agencies, e.g. EUSPA, EEA, FRONTEX, SATCEN, EMSA, EDA

Besides the above mentioned Delegation and Contribution agreements, similar agreements have been concluded with the **European Agency for the Space Programme (EUSPA)** in the area of GNSS exploitation activities. In 2022, the EU Agency for the Space Programme (EUSPA) constituted the biggest part of DG DEFIS expenditure, totalling EUR 1.091 billion (41,50 % of the total DG DEFIS 2022 payments). Although DG DEFIS remains ultimately accountable for the legality and regularity of these expenditures, it is the responsibility of to set up the appropriate controls to provide the necessary assurance to its parent Directorate-General. The Commission exercises the supervisory tasks provided for in the existing delegation agreements through ex-ante controls over the regular implementation reports and procurement documentation submitted by the Agency. EUSPA is audited annually by the European Court of Auditors (ECA) and the submitted AIRs are also audited by external independent auditors. DG DEFIS ex-post control team audited all annual AIRs submitted by ESA/EUSPA. The overall detected error rate for Galileo and EGNOS for 2022 following the DG DEFIS audits (Working Arrangement (WA) with ESA included) is 0.06 % (respectively 0.01 % for GALILEO and 0.30% for EGNOS). Given that the grants provided by EUSPA under the H2020 delegation agreement are audited by the Common Audit Service (CAS) of the Research family, DG DEFIS used the related error rates (Representative error rate of 2.71% and cumulative residual error rate of 1.67 %).

Result indicators: Indicators of annual error – GALILEO and EGNOS programmes

GALILEO						
Annual Implementation Report of Financial Year	Reported by GSA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2016	19.488.475	18.628.058	-860.417	4.42 %	860 417	0
2017	432.571.139	432.518.462	-52.677	0.01 %	-52 677	0
2018	480.850.890	480.903.567	52.677	-0,01%	52.677	0
2019	478.214.483	477.916.815	297.668	0.1%	297.668	0
2020	650.661.073	650.704.929	43.857	0,01%	43.857	0
2021	923.972.662	924.058.371	85.709	0,01%	85.709	0

EGNOS						
Annual Implementation Report of Financial Year	Reported by GSA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2016	98.028.677	95.625.748	-2.402.929	2.45 %	2 402 929	0
2017	103.969.092	103.464.571	-504.521	0.49 %	504 521	0
2018	159.776.396	160.169.764	393.368	-0.24%	393.368	0
2019	192.642.229	190.916.126	-1.726.102	1.29%	-1.726.102	0
2020	121.735.489	122.144.522	409.034	0,57%	409.034	0
2021	139.059.958	139.374.969	315.011	0,30%	315.011	0

According to the DG DEFIS audit guidelines and the 2022 Audit Work Programme, the entities other than ESA and EUSPA are audited every two years. However, the **EEA** has been audited again in 2022 since risks and errors were identified during previous years. The audit, carried out on the 2021 Annual Implementation Report, revealed no errors.

Result indicators: Indicators of annual error – EEA

Annual Implementation Report of Financial Year	Reported by EEA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2016	4.871.854	4.691.072	-180.782	3,71%		
2017	not audited					
2018	18.407.180	16.912.013	-1.495.167	11.99%		
2019	12.118.540	12.251.048	132.508	1.23%		
2020	20.659.705	20.991.511	331.806	2,41%	331.806	0
2021	17.391.384	17.391.384	0	0,00%	0	0

FRONTEX

In line with the Annual Audit Work Programme, an audit was performed on the 2021 costs reported. The Audit did not reveal any errors. Considering that DG DEFIS is not the parent DG of FRONTEX, its supervision activities are focused on the implementation of the tasks delegated to the Agency under the signed Delegation Agreement (DA) and the Contribution Agreement (CA).

The DA/CA ensures that the Commission, the European Anti-fraud Office and the Court of Auditors or their authorised representatives, may at any time during the implementation of the entrusted tasks and up to five years after the payment of the balance carry out checks and audits on the implementation of the entrusted tasks.

Result indicators: Indicators of annual error – FRONTEX

Annual Implementation Report of Financial Year	Reported by FRONTEX	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2017	6.505.198	7.875.846	1.370.648	0,00%		
2018	9.723.273	9.746.664	23.391	-0,25%		
2019	8.369.994	8.364.342	5.652	0.07%		
2020	not audited					
2021	8.107.753	8.082.753	-25.000	0,31%		

In 2022, audits were performed in line with the bi-annual control foreseen for both EMSA and SatCen in the DG DEFIS Control Strategy.

EMSA						
Annual Implementation Report of Financial Year	Reported by EMSA	Commission Audit report	Adjustment	Detected error rate	Implemented amount via clearing of pre-financing	Amount to be implemented
2017	6.421.992	6.076.446	-345.546	0	345.546	
2018	not audited					
2019	9.378.446	9.382.383	3.937	-0.05%		
2020	not audited					
2021	6.339.063	6.313.485	-25.578	0,44%		

SATCEN						
Annual Implementation Report of Financial Year	*Reported by SatCen*	*Commission Audit report*	*Adjustment*	*Detected error rate*	*Implemented amount via clearing of pre-financing*	*Amount to be implemented*
2017	1.969.701	1.985.199	15.498	0,00%		
2018	Not audited					
2019	6.078.214	6.078.214	0	0,00%		
2020	Not audited					
2021	6.375.332	6.325.965	-49.367	1,61%		

No audits have been carried out in 2022 on **EDA** as only pre-financing were paid in 2021.

Although the detected error rates during the ex-post control in 2022 is very low, all decentralised agencies were given a “theoretical” error rate of 0.5%, following the conservative approach mentioned above. It is to be mentioned that according to DG DEFIS audit guidelines, these Agencies are audited every 2nd year, unless a specific risk is raised. Ex-ante checks and regular monitoring is performed on these entities over the year to ensure legality and regularity of the transactions.

Based on the declarations of assurance provided by the Executive Directors of these decentralised agencies and the results of the performed ex-post audits, DG DEFIS considers the implementation of the delegated funds to be legal and regular.

Executive Agencies

In its capacity of parent DG, DG DEFIS supervises the control systems[38] of REA in the context of their direct delegation as Authorising Officer by Delegation (AOD). REA performed its ex-post audits in the context of a common audit strategy. REA produces its own AARs. The preparation of the Annual Activity Report of REA was supervised by DG DEFIS and by the Steering Committee of the Agency.

Overall, DG DEFIS considers that its supervision of REA is effective and appropriate.

([38]) The control systems of the Executive Agencies are similar to those of their parent DGs.

C. Procurement

Procurement under direct management represents 0.75 % of the total 2022 DG DEFIS expenditure. The payments made in 2022 on own procurement amount to EUR 19.788.982 million.

The Relevant Control System template (RCS) n°3 for procurement in Annex 6 demonstrates how the control system in place in the Directorate-General addresses the risks related to this type of expenditure.

In 2022, 15 contracts with a value exceeding EUR 60,000 were awarded directly by DG DEFIS, representing a total contract value of EUR 420.4 million. The reader is referred to Annex 3, table 12. These contracts do not include, however, contracts signed by the ESA in the name and on behalf of the Commission under ESA Delegation Agreements but include a large framework contract launched and awarded by the Commission for the provision of launch services for Copernicus Sentinel satellites that will be implemented by ESA in indirect management.

The procurement procedures applied in DG DEFIS involve a number of specific controls, which are fully in line with the applicable regulatory requirements. The benefit of these specific controls provides assurance on legality and regularity, transparency, equal treatment and proportionality of the public procurement and mitigates the risk of reputational damage. Given the low error rate, there are no indication that a higher level of checks and controls would produce any supplementary benefits.

D. Grants directly managed by DG DEFIS

DG DEFIS has set up internal control processes aimed to ensure the adequate management of the risks relating to the legality and regularity of the underlying transactions, taking into account the specificities of the programmes, as well as the nature of the payments concerned.

The major part of the expenditure implemented in 2022 in direct management relates to the European Defence Fund (EDF). As result of the 2021 EDF call for proposals 56 grant agreements were signed for a total value of EUR 983,7M. The pre-financing paid for these grants amounts to EUR 439M. For the grants funded under European Defence Industrial Development Programme (EDIDP) an amount of EUR 5 M was used in interim payments.

No ex-post audits have yet been carried out over these grants, as the first EDIDP projects started at the end of 2020 and have an average duration of 3 years.

E. Audits from the Internal Audit Service (IAS) and the European Court of Auditors (ECA)

During the first three years of its existence, DG DEFIS was subject to several audits both from the Internal Audit Service (IAS) and from the European Court of Auditors (ECA), covering various operational, financial, and organisational aspects of its defence industry and space programmes. The space activities were already regularly audited before the creation of DG DEFIS.

In the last five years (2018-2022), the IAS conducted eight audits, including one still in progress:

- Audit on the supervision of project management and payment for Galileo (DG GROW, 2018).
- Audit on intellectual property rights supporting activities (DG GROW, 2018).
- Audit on the supervision of the implementation of the Copernicus Programme - Phase 2: management of the cooperation and coordination (DG GROW, 2018).
- Audit on the supervision of the implementation of the 2014-2020 programme for the European Geostationary Navigation Overlay Service (EGNOS) (2020).
- Audit on payments and accounting for tangible assets under the Galileo and Copernicus 2014-2020 programmes (2020).
- Limited review on the error rate calculation (2021).
- Audit on the preparedness for the management of the European Defence Fund (2022 – concluded early 2023).
- Audit on the preparedness of DG DEFIS's management and control systems for the 2021-2027 Space programme implementation (started in 2022 and still ongoing).

Based on the work done, the IAS concluded that the internal control systems in place for the audited processes were effective, except very recently for the observation giving rise to a "very important recommendation" concerning the validation of the SME (or Midcap) status of beneficiaries, in the context of the European Defence Fund calls.

DG DEFIS Management accepted all the recommendation and submitted specific action plans for their implementation. The IAS has considered all these action plans as adequate to address the residual risks identified by the auditors. The IAS closely monitors their implementation through follow-up reports and audits.

The European Court of Auditors conducted, in addition to its ordinary annual audits on accounts, two performance audits on DG DEFIS, one of which is reaching its final stage in 2023. These audits concerned the following subjects:

- Special Report 07/2021: EU space programme Galileo and Copernicus: services launched, but the uptake needs a further boost.
- Draft Special Report: The Preparatory action on defence research: Some lessons learned, but value as a testbed for increasing EU defence spending reduced due to time constraints and limited results .

As far as the Special Report 07/2021 is concerned, DG DEFIS Management accepted all the issued recommendations. Their implementation is ongoing, with first milestones foreseen by the end of 2023.

1. Table Y on the estimated "cost of controls" at Commission level

Table Y Cost of control

Relevant Control System	Ex ante controls			Ex post controls			Total	
	a	b	c	d	e	f	g	h
	EC total costs	funds managed (in EUR)*	Ratio (%)* a/b	EC total costs	total value verified and/or audited (in EUR)	Ratio (%) d/e	EC total estimated cost of controls (in EUR)	Ratio (%) g/b
RCS 1 : Budget entrusted to other entities	4.005.966	2.145.922.747	0,19%	210.630	1.485.158.195	0,01%	4.216.596	0,20%
RCS 3: Procuremen	1.329.850	25.892.536	5,14%	0	0		1.329.850	5,14%
RCS 4: Grants	6.157.918	456.049.502	1,35%		0		6.157.918	1,35%
OVERALL estimated cost of control at EC level	11.493.734	2.627.864.785	0,44%	210.630	1.485.158.195	0,01%	11.704.364	0,45%

Cost of control heading 1

Relevant Control System	Ex ante controls			Ex post controls			Total	
	a	b	c	d	e	f	g	h
	EC total costs	funds managed (in EUR)*	Ratio (%)* a/b	EC total costs	total value verified and/or audited (in EUR)	Ratio (%) d/e	EC total estimated cost of controls (in EUR)	Ratio (%) g/b
RCS 1 : Budget entrusted to other entities	3.520.841	2.072.054.995	0,17%	210.630	1.485.158.195	0,01%	3.731.471	0,18%
RCS 3: Procuremen	1.252.770	23.018.353	5,44%	0	0		1.252.770	5,44%
RCS 4: Grants	259.100	12.049.324	2,15%	0	0		259.100	2,15%
OVERALL estimated cost of control at EC level	5.032.711	2.107.122.672	0,24%	210.630	1.485.158.195	0,01%	5.243.341	0,25%

Cost of control heading 5

Relevant Control System	Ex ante controls			Ex post controls			Total	
	a	b	c	d	e	f	g	h
	EC total costs	funds managed (in EUR)*	Ratio (%)* a/b	EC total costs	total value verified and/or audited (in EUR)	Ratio (%) d/e	EC total estimated cost of controls (in EUR)	Ratio (%) g/b
RCS 1 : Budget entrusted to other entities	485.125	73.867.752	0,66%	0	0		485.125	0,66%
RCS 3: Procuremen	77.080	2.874.183	2,68%		0		77.080	2,68%
RCS 4: Grants	5.898.818	444.000.178	1,33%	0	0		5.898.818	1,33%
OVERALL estimated cost of control at EC level	6.461.023	520.742.113	1,24%	0	0		6.461.023	1,24%

The audit coverage for FP7 is the same as the one presented in the AAR 2021 because all Common Representative sample items were closed in 2021.

Horizon 2020 (H2020)

By the end of 2022, **the Research and Innovation Family** audited 4 060 participations, covering 58.26% of total H2020 expenditure to date.

The percentage of H2020 expenditure covered by the audits (58.26%) refers to the value of the participations of the audited beneficiaries. It includes both fully audited participations (3.77%), also referred to as the 'direct' coverage, and the non-audited participations, also referred to as the 'indirect' coverage, which after the full treatment of audit results, are clean from systemic errors (54.49%).

Completion rate of Horizon 2020 ex-post audits in 2022

The overall target (most probable scenario) in the Horizon 2020 Audit Strategy for 2022 was 611 audited participations. By 31 December 2022, the audits of 633 participations were closed, (completion rate 103.6%).

Results of the research programmes ex-post audits Seventh Framework programme, Horizon 2020 an Horizon Europe)

Seventh Framework Programme

The audit strategy for the Seventh Framework Programme was already considered to be fully implemented in 2020[39]. As was the case last year, because of the 'de minimis'

[39] See section on Seventh Framework Programme in AAR 2021 (Representative detected error rate : 5.44% and Cumulative residual error rate: 3.47% for DG Research and Innovation)

threshold for financial reservations introduced in 2019[40], a quantified reservation is not required for the Seventh Framework Programme for Research and Innovation[41].

Horizon 2020 Framework Programme

In 2020, the Commission refined its methodology for calculating the Horizon 2020 error rates in line with the European Court of Auditors' observations in its 2018 and 2019 Annual Reports[42]. The methodology applied is described in Annex 5 'Materiality criteria'. As of January 2020, DG R&I applied the revised methodology on a sample of 1 937 audit conclusions. This results in the following error rates for Horizon 2020[43] on 31 December 2022:

- Cumulative representative detected error rate: **2.71%**[44]
- Cumulative residual error rate for the Research and Innovation Family DGs: **1.67**% (1.71 % for DG Research and Innovation[45]).

In line with the Financial Statement[46] accompanying the Commission's proposal for the Horizon 2020 regulation, a reservation is not necessary for the related expenditure if the cumulative residual error rate for the programme falls within the target range of 2-5%. In 2022, and despite the above-mentioned caveats, DG R&I cumulative residual error rate for Horizon 2020, calculated at 1.71%, more than fulfils this condition and is below the materiality threshold. Despite the absence of reservation, the root causes of errors have been identified and targeted actions taken to address any identified weaknesses.

---

[40] Agreement of the Corporate Management Board of 30/4/2019.

[41] In 2019 , a 'de minimis' threshold for financial reservations was introduced stipulating that quantified Annual Activity Report reservations related to residual error rates above the 2% materiality threshold are deemed not substantial for segments representing less than 5% of a DG's total payments and with a financial impact below EUR 5 million. As is the case for the Seventh Framework Programme expenditure , a quantified reservation is not required. FP7 payments represent 0.23% of 2022 R&I payments and the financial impact is EUR 0.66 million.

[42] When calculating the multi-annual error rate, the Commission took into account the results of the audit re-performed by the ECA as part of Module 2 of the DAS 2018-2019.

[43] The Horizon 2020 audit campaign started in 2016. At this stage, four Common Representative Samples with a total of 628 expected results have been selected. By the end of 2022, cost claims amounting to EUR 40.8 billion have been submitted by the beneficiaries to the services. The audit coverage for Horizon 2020 is presented in annex 7. In addition to the Common Representative Samples, Common Risk Samples and Additional Samples have also been selected. The audits of 4 060 participations were finalised by 31/12/2022 (of which 633 in 2022).

[44] Based on the 479 representative results out of the 628 expected in the four Common Representative Samples.

[45] It should be noted that in 2021 most H2020 grants managed by DG R&I were transferred to Executive Agencies. Hence, this figure is based only on the actions that remained with DG R&I at the end of 2021.

[46] The legislative financial statement accompanying the Commission's proposal for the Horizon 2020 regulation states: "The Commission considers therefore that, for research spending under Horizon 2020, a risk of error, on an annual basis, within a range between 2-5% is a realistic objective taking into account the costs of controls, the simplification measures proposed to reduce the complexity of rules and the related inherent risk associated to the reimbursement of costs of the research projects. The ultimate aim for the residual level of error at the closure of the programmes after the financial impact of all audits, corrections and recovery measures will have been taken into account is to achieve a level as close as possible to 2%."

Since Horizon 2020 is a multi-annual programme, the error rates, and the residual error rate in particular, should be considered within a time perspective. Specifically, the cleaning effect of audits will tend to increase the difference between the representative detected error rate and the cumulative residual error rate, with the latter finishing at a lower value.

These error rates are calculated on the basis of the audit results available when drafting the Annual Activity Report. They should be treated with caution as they may change subject to the availability of additional data from audit results.

Given the results of the audit campaign, and the observations made by the European Court of Auditors in its Annual Reports, the Common Implementation Centre, in close cooperation with central Commission services, defined actions aimed at significantly simplifying the rules, and paving the way for a significant reduction of the error rate in Horizon Europe. Apart from the use of a Corporate Model Grant Agreement and a common Annotated Grant Agreement for all programmes directly managed by the Commission, other actions include further simplification, such as the increased use of simplified forms of funding (including lump sums and unit costs), focused communication campaigns to more "error-prone" types of beneficiaries with higher than average error rates, such as SMEs and newcomers, and enhanced training to external audit firms performing audits on behalf of the Commission (the last three measures also target H2020 grants and beneficiaries). Focusing on the most common errors, these events will be straightforward, reaching more participants and achieving higher impact.

In addition, the ECA recommended certain improvements related to the quality of the audit process. Following the recommendation of the ECA the Commission reinforced or introduced a number of actions to remedy the risks identified by the ECA. The Court has acknowledged all these efforts to improve the quality of audits and considers its recommendation to be fully implemented.

Horizon Europe Framework Programme

2022 was the second year of implementation of the Horizon Europe framework programme. No representative error rate for Horizon Europe is available in 2022 as the ex-post audit campaign for the Programme is planned to be launched by the end of 2023 at the earliest, once a meaningful number of payments can be audited. Consequently, without elements allowing an assessment of the level of errors and taking into account the simplifications introduced for Horizon Europe that aim to reduce the error rate, the detected and residual error rate will be estimated to 2%[47] for DG R&I.

[47] These rates corespond to the objective set for Horizon Europe.

ANNEX 8: Specific annexes related to "assessment of the effectiveness of the internal control systems"

N/A

ANNEX 9: Specific annexes related to "Control results" and “Assurance: Reservations”

DG DEFIS - Estimated risk at payment and at closure

DG DEFIS -1	Payments made (2022;MEUR) -2	minus new prefinancing [plus retentions made] (in 2022;MEUR) -3	plus cleared prefinancing [minus retentions released and deductions of expenditure made by MS] (in 2022;MEUR) -4	Relevant expenditure (for 2022;MEUR) -5	Detected error rate or equivalent estimates -6	Estimated risk at payment (2022;MEUR) -7	Adjusted Average Recoveries and Corrections (*adjusted* ARC; %) -8	Estimated future corrections [and deductions] (for 2022;MEUR) -9	Estimated risk at Closure (2022;MEUR) -10
Administrative expenses	6,10	0,00	0,62	6,72	0,50% - 0,50%	0,03 - 0,03	0,00% - 0,00%	0,00 - 0,00	0,03 - 0,03
Heading 1 -Own procurement	17,58	- 2,78	2,90	17,70	0,50% - 0,50%	0,09 - 0,09	0,00% - 0,00%	0,00 - 0,00	0,09 - 0,09
Heading 1 -Own grants	12,05	- 3,99	21,77	29,82	0,50% - 0,50%	0,15 - 0,15	0,00% - 0,00%	0,00 - 0,00	0,15 - 0,15
Heading 5 -EDF Grants	444,00	- 439,02	0,00	4,98	0,50% - 0,50%	0,02 - 0,02	0,00% - 0,00%	0,00 - 0,00	0,02 - 0,02
Heading 5 -EDF Procurement	2,21	0,00	0,00	2,21	0,50% - 0,50%	0,01 - 0,01	0,00% - 0,00%	0,00 - 0,00	0,01 - 0,01
Heading 1 -International Organisations -ESA - Copernicus	657,85	- 657,85	361,10	361,10	0,50% - 0,50%	1,81 - 1,81	0,00% - 0,00%	0,00 - 0,00	1,80 - 1,80
Heading 1 -International Organisations -ESA - IOV/IOD	8,00	- 8,00	6,09	6,09	0,50% - 0,50%	0,03 - 0,03	0,00% - 0,00%	0,00 - 0,00	0,03 - 0,03
Heading 1 -International Organisations -ESA - GALILEO	58,16	- 58,16	465,82	465,82	0,50% - 0,50%	2,33 - 2,33	0,00% - 0,00%	0,00 - 0,00	2,33 - 2,33
Heading 1 -International Organisations -ESA - Govsatcom/SSA	6,91	- 6,91	1,54	1,54	0,50% - 0,50%	0,01 - 0,01	0,00% - 0,00%	0,00 - 0,00	0,01 - 0,01
Heading 1 -International Organisations -ESA - Horizon 2020/Europe	22,29	- 22,29	24,20	24,20	0,50% - 0,50%	0,12 - 0,12	0,00% - 0,00%	0,00 - 0,00	0,12 - 0,12
Heading 1 -International Organisations -Copernicus - Mercator	25,21	- 25,21	28,48	28,48	0,50% - 0,50%	0,14 - 0,14	0,00% - 0,00%	0,00 - 0,00	0,14 - 0,14
Heading 1 -International Organisations -Copernicus - ECMWF	22,56	- 22,56	48,54	48,54	0,50% - 0,50%	0,24 - 0,24	0,00% - 0,00%	0,00 - 0,00	0,24 - 0,24
Heading 1 -International Organisations -Copernicus - EUMETSAT	81,03	- 81,03	64,81	64,81	0,50% - 0,50%	0,32 - 0,32	0,00% - 0,00%	0,00 - 0,00	0,32 - 0,32
Heading 1 -International Organisations -Copernicus - EEA	20,74	- 20,74	17,39	17,39	0,50% - 0,50%	0,09 - 0,09	0,00% - 0,00%	0,00 - 0,00	0,09 - 0,09
International Organisations -Copernicus - EMSA	12,88	- 12,88	6,31	6,31	0,44% - 0,44%	0,03 - 0,03	0,00% - 0,00%	0,00 - 0,00	0,03 - 0,03
Heading 1 -International Organisations -Copernicus - SatCen	0,35	- 0,35	6,33	6,33	1,61% - 1,61%	0,10 - 0,10	0,00% - 0,00%	0,00 - 0,00	0,10 - 0,10
Heading 1 -International Organisations - Copernicus - Frontex	0,00	0,00	8,08	8,08	0,31% - 0,31%	0,03 - 0,03	0,00% - 0,00%	0,00 - 0,00	0,03 - 0,03
Heading 5 -International Organisations -OCCAR	57,04	- 52,76	0,00	4,29	0,50% - 0,50%	0,02 - 0,02	0,00% - 0,00%	0,00 - 0,00	0,02 - 0,02
Heading 1 -Agencies -EUSPA Subsidy	65,39	- 65,39	44,14	44,14	0,50% - 0,50%	0,22 - 0,22	0,00% - 0,00%	0,00 - 0,00	0,22 - 0,22
Heading 1 -EUSPA-EGNOS	78,65	- 78,65	139,30	139,30	0,30% - 0,30%	0,42 - 0,42	0,00% - 0,00%	0,00 - 0,00	0,42 - 0,42
Heading 1 -EUSPA- GALILEO	1 000,54	-1 000,54	925,22	925,22	0,01% - 0,01%	0,09 - 0,09	0,00% - 0,00%	0,00 - 0,00	0,09 - 0,09
Heading 1 -EUSPA - Govsatcom/SSA	3,14	- 3,14	0,26	0,26	0,50% - 0,50%	0,00 - 0,00	0,00% - 0,00%	0,00 - 0,00	0,00 - 0,00
Heading 1 -EUSPA - H2020/Europe - downstream	8,36	- 8,36	16,38	16,38	2,71% - 2,71%	0,44 - 0,44	0,00% - 0,00%	0,00 - 0,00	0,44 - 0,44
EDA	16,82	- 16,82	18,67	18,67	0,50% - 0,50%	0,09 - 0,09	0,00% - 0,00%	0,00 - 0,00	0,09 - 0,09
DG total	**2 627,86**	**-2 587,42**	**2 207,95**	**2 248,39**		**6,84 - 6,84**	**0,00% - 0,00%**	**0,00 - 0,00**	**6,84 - 6,84**
					Overall risk at payment in %	**0,30% - 0,30%** *(7) / (5)*		**Overall risk at closure in %**	**0,30% - 0,30%** *(10) / (5)*

Notes to the table X

(1) Relevant Control Systems differentiated per relevant portfolio segments and at a level which is lower than the total.

(2) Payments made or equivalent, e.g. expenditure registered in the Commission’s accounting system, accepted expenditure or cleared pre-financing. In any case, this means after the preventive (ex-ante) control measures have already been implemented earlier in the cycle.

In all cases of Co-Delegations (Internal Rules Article 3), "payments made" are reported by the Delegated departments. For Cross-Sub-Delegations (Internal Rules Article 12), the reporting remains with the Delegating departments.

(3) New pre-financing actually paid out by the department itself during the financial year (i.e. excluding any pre-financing received as a transfer from another department). as per note 2.5.1 to the Commission annual accounts thus excluding "Other advances to Member States" which are covered on a purely payment-made basis (note 2.5.2). Pre-financing paid/cleared" are always covered by the Delegated departments, even for Cross-Sub-Delegations.

(4) Pre-financing actually cleared during the financial year (i.e. their 'delta' in the Financial Year 'actuals', not their 'cut-off' based estimated 'consumption').

(5) For the purpose of equivalence with the ECA's scope of the EC funds with potential exposure to legality & regularity errors (see the ECA's Annual Report methodological annex 1.1), our concept of "relevant expenditure" includes the payments made, subtracts the new pre-financing paid out [& adds the retentions made], and adds the pre-financing actually

cleared [& subtracts the retentions released; and any deductions of *expenditure made by MS*] during the FY. This is a separate and 'hybrid' concept, intentionally combining elements from the budgetary accounting and from the general ledger accounting.

(6) In this column, we disclose the detected error rates or equivalent estimates

For low-risk types of expenditure, where there are indications that the equivalent error rate might be close to 'zero' (*e.g. administrative expenditure, operating contributions to agencies*), the rate which should be used is 0.5% as a conservative estimate, unless the department has a more precise estimate based on evidence.

(8) The adjusted average recovery and corrections percentage is based on the 7 years historic Average of Recoveries and financial Corrections (ARC), which is the best available indication of the corrective measures each department applied over the past years as a result of ex post controls. The AOD has not adjusted this historic average.

DG DEFIS has analysed the correlation between estimated future corrections and implemented amount of corrections and recoveries. However, since the estimated future corrections over the years declines from 170 K EUR to 3K EUR and as we work in multi-year programmes, where the corrections are implemented in the next reporting year and hardly result in real ROs, no trend can be established.

(9) For some programmes with no set *closure* point (e.g. EAGF) and for some multiannual programmes for which corrections are still possible afterwards (e.g. EAFRD and ESIF), all corrections that remain possible are considered for this estimate.

ANNEX 10: Reporting – Human resources, digital transformation and information management and sound environmental management

Human resources management

Objective: DG DEFIS employs a competent and engaged workforce and contributes to gender equality at all levels of management to effectively deliver on the Commission's priorities and core business		
Indicator 1: Number and percentage of first female appointments to middle management positions **Source of data:** Commission Decision SEC(2020)146 of 1 April 2020 and SYSPER		
Baseline **(2020)**	**Target**[48] **(2022) + (2024)** **N**	**Latest known results** (2022)
Female representation in management: 56%, 5 out of 9 on 1 February 2020	1 female Head of Unit appointment	1 female Head of Unit appointment out of 3 in total in 2021 (33%). 2 female appointments since target set. 1 more female nomination in the middle management in 2022 (50% female Heads of Unit)
Indicator 2: DG DEFIS Staff engagement index **Source of data:** Commission staff survey		
Baseline **(2020)**	**Target** **(2024)**	**Latest known results** (2022)
60%[49]	70%	66%

(48) The target will be reviewed for the period 2023-2024 by January 2023.

(49) DG DEFIS was created end of 2019 by the transfer from DG GROW of Directorates I and J. The staff engagement index for these two directorates was 60% in 2018 and was taken as baseline for DG DEFIS.

Main outputs in 2022:			
Human resources management			
Description	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Promote virtual and e-learning opportunities for colleagues (i.e. webinars, e-learning modules)	Number of virtual and e-learning trainings followed by DEFIS staff	10 % of trainings followed by virtual means	11% In 2022, DEFIS staff participated in 1107 courses, of which 118 were e-learning courses.
Launch specialised EPSO competitions	Official publication	1 competition completed in 2022 in defence and space domain	The competition for defence and space domains launched in June 2022 by EPSO. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ%3AC%3A2022%3A233A%3ATOC
Issue local HR strategy	Publication	Q2 2022	Done in 2022 via Local HR strategy template: Workforce planning – learning priorities
Organise an away day	Staff attendance	90 % of DG DEFIS staff attendance (if the sanitary conditions in 2022 allow)	Almost all DG DEFIS staff attended (the target of 90% slightly exceeded)

Internal communication actions			
Output	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Maintenance and animation of DG DEFIS intracomm website	- Number of visits	- 5.000 visits	- 36.799 visits
Publication of weekly internal newsletters 'DEFIS BUZZ'	- Number of recipients and readers	- 150 readers per edition -	- 296 readers per edition
Organisation of DEFIS Lab seminars	- Number of editions - Number of participants	- 6 editions - 400 participants	- 3 editions - 240 participants
Organisation of a DG DEFIS Away-day	- Number of participants and satisfaction survey	- At least 60% of participants	- 69% of participants
Organisation of regular meeting(s) of the Director-General with the staff	- Number of meetings - Number of participants and quality of interactions	- 4 meetings - 175 participants each	- 5 meetings - 225 participants per meeting
Promotion of EMAS and organisation of EMAS events to promote corporate campaigns	- Number of events	- At least 1	- Regular promotion of EMAS initiatives - DEFIS actions on emissions reduction and energy savings

Digital Transformation and information management

DG DEFIS responsibilities under this area were to a large extent shared with the Directorate-General Internal Market, Industry, Entrepreneurship and SMEs (DG GROW). A Memorandum of Understanding (MoU) was signed between DG DEFIS and DG GROW, which expired on the 31st of December 2022.

DG DEFIS ensured the business continuity with a smooth transition considering the most efficient use of existing staff resources, as well as the necessary synergies and efficiency in terms of strategic horizontal services.

Objective: DG DEFIS is using innovative, trusted digital solutions for better policy-shaping, information management and administrative processes to forge a truly digitally transformed, user-focused and data-driven Commission		
Indicator 1: Degree of implementation of the digital solutions modernisation plan[50] **Source of data:** Commission Decision SEC(2020)146 of 1 April 2020 and SYSPER		
Baseline (2020)	**Target (2024)**	**Latest known results (Situation on 31/12/2022)**
0%	60 %	50 %

Indicator 2: Percentage of DG DEFIS key data assets for which corporate principles for data governance have been implemented **Source of data:** DG DEFIS			
Baseline (2020)	**Interim milestone** (2022)	**Target (2024)**	**Latest known results** (situation on 31/12/2022)
0%[51]	25%	50%	25%

Indicator 3: Percentage of staff attending awareness raising activities on data protection compliance **Source of data:** DG DEFIS		
Baseline (2020)	**Target (2024)**	**Latest known results** (situation on 31/12/2022)
0% (DG DEFIS created on 1/1/2020)	100% of staff	54% overall staff awareness level reached in DEFIS on the basis of DPC-led activities since 2020. 4 entities/units trained in 2022, 3 in 2021, and 7 out of 13 entities/units trained in total since 2020.

(50) The European Commission Digital Strategy (C(2018)7118) calls on Commission services to digitally transform their business processes by developing new innovative digital solutions or make evolve the existing ones in line with the principles of the strategy. At the beginning of the year N+1, the Solution Owner and IT Investments Team will assess the progress made on the basis of the proposed modernisation plan. For each of the 3 solutions, a table will reflect – per principle - the progress achieved during the last year.

(51) DG DEFIS did not complete a statistics and data inventory in 2020, as it has just been created on 1 January 2020

Main outputs in 2022: Digital transformation and information management			
Description	**Indicator**	**Target**	**Latest known results** (Situation on 31/12/2022)
Implementation of the corporate principles for data governance for DG DEFIS key data assets	Percentage of implementation of the corporate principles for data governance for [the service's] key data assets	Interim milestone by 2022: 50%	25 %
Deployment of new IT applications and update of existing application (listed in the Management Plan 2022)	Number of new DEFIS IT application	3 new or update of existing applications	2
Increase staff awareness in DG DEFIS on personal data protection rules.	- Percentage management attending awareness raising activities. - Percentage of staff attending awareness raising activities.	- Senior Management: 50% - Middle Management: 50% - Other Staff: 50%	- 61 % (8 units out of 13) reached during a dedicated mandatory training session targeted at middle managers. - 100% of DEFIS staff was reached during the data protection day via a mailing. - 20 % of staff (55 out 281 reached by internal trainings by the DPC. - 26% of all staff (73 of 281 by Dec 2022).
Map degree of implementation of data protection procedures	- Number of records due. - Number of privacy statements due. Percentage of Unit inventories	- Percentage of records prepared. - Percentage of privacy statements prepared.Number of Unit inventories prepared.	- 18 % (2 out of 11) with 2 published records 9 records initiated (in draft) during 2022 which are pending for finalisation in 2023. - 100% with 45 privacy statements validated by the DPC.

Description	Indicator	Target	Latest known results (Situation on 31/12/2022)
DEFIS IT systems complying with data protection rules.	Number of DEFIS IT systems.	Percentage of DEFIS IT systems complying with data protection rules.	- 27% (3 out of 11)

Sound environmental management and example(s) of initiatives to improve economy and efficiency of financial and non-financial activities

Objective: DG DEFIS takes full account of its environmental impact in all its actions and actively promotes measures to reduce the related day-to-day impact of the administration and its work			
Main outputs in 2022:			
I. More efficient use of resources (energy, water, paper):			
Description	**Indicator**	**Target (2019 as baseline)**	**Latest known results** (situation on 31/12/2022)
Paperless working methods at DG level (such as paperless working: e-signatories, financial circuits, collaborative working tools) and staff awareness actions to reduce office paper use in the framework of EMAS corporate campaigns and/or raise awareness about DG's office paper use in collaboration with OIB/OIL where appropriate.	Number of actions	1 action per year	
	Number or % of staff informed/participated	Address all DG DEFIS staff	All staff informed
	% of Ares e-signed documents	establish a baseline for the year of operation	
		Reduce paper consumption (%)	
II. Reducing CO_2, equivalent CO_2 and other atmospheric emissions			
Description output	**Indicator**	**Target (2019 as baseline)**	**Latest known results** (situation on 31/12/2022)
Staff awareness actions on reducing GHG emissions (such as actions on sustainable commuting	Number or % of staff informed/participated	Address all DG staff	All staff was informed
	% of staff participating in VeloWalk	increase of staff participating in VeloWalk	

Description output	Indicator	Target (2019 as baseline)	Latest known results (situation on 31/12/2022)
during EU Mobility week and VeloWalk corporate events) and/or raise staff awareness on sustainable commuting in collaboration with OIB or OIL (e.g. availability of bike parking facilities, lockers and showers, promote the reduction of parking spaces' use amongst staff).	% of sustainable commuters ([52]) at DG/service	Increase % of sustainable commuters at DG/service (%) ([53])	
Gradual increased use (and number of) VC ([54]) meeting rooms for meetings with stakeholders (avoiding business trips) in the DG, in collaboration with DG SCIC, OIB and OIL.	Number of VC / hybrid meeting rooms	Number of VC /Hybrid meeting rooms	
Analysis of DG's missions trends /patterns (based on corporate EC-staff's professional trips (missions) ([55])), optimise and gradually reduce CO_2 emissions (e.g. by optimising the number of participants in the same mission, promoting more sustainable travelling options, promoting videoconferencing/ virtual events as an alternative).	Number of missions	Number of missions	
	CO_2 (t) emissions from DG's missions *(% means of transportation used)*	Establish a baseline for the first "normal" year of operation Reduce DG's CO_2 emissions from missions ([56]) (%) in comparison with 2019 data (while in DG GROW)	
Staff awareness on digital pollution and gradual change of behaviours avoiding heavy emails, encouraging the use of ICT platforms, avoiding unnecessary storage of data.	Number or % of staff informed/participated	Address all DG staff	All staff was informed/reminded through internal communication

([52]) Sustainable commuting usually refers to environmentally friendly travel modes, such as. Public transport (bus, tram, subway, light rail), walking, cycling, and carpooling.
([53]) Only for Brussels: Based on the results of the staff mobility surveys conducted by OIB.
([54]) VC (Videoconferencing) room
([55]) data provided by HR.D.02
([56]) Based on data provided by the Commission's carbon footprint analysis (by HR.D.02), including business travel.

III. Reducing and management of waste			
Description output	**Indicator**	**Target (2019 as baseline)**	**Latest known results** (situation on 31/12/2022)
Implementation of the EC Guidelines for sustainable meetings and events, e.g. reduce/eliminate single-use plastics, gadgets/gifts.	Number of green events	At least 1 event	8 events under the Green Deal objective

IV. Promoting green public procurement (GPP)			
Description output	**Indicator**	**Target (2019 as baseline)**	**Latest known results** (situation on 31/12/2022)
Encourage actions underpinning the use of space data, services and applications in support of the EU Green Deal.	Number of calls	Equal or above 2021 results	
	Number of projects funded	Equal or above 2021 results	
	Overall amount in EUR	Equal or above 2021 results	

Security and Information

Objective: DG DEFIS is ensuring a high level of protection of the sensitive and classified information it manages		
Indicator 1: Percentage of staff holding a Personal Security Clearance **Source of data:** LSO database		
Baseline **(2020)** **DG DEFIS created on 1/1/2020**	**Target** (2024)	**Latest known results** (situation on 31/12/2022)
95%	100%	81.5% holding a Personal Security Clearance; 16% under vetting procedure

Indicator 2: Number of security incidents detected versus number of transfers of EUCI (on a yearly basis) **Source of data:** LSO database, RCO database		
Baseline **(2020)** **DG DEFIS created on 1/1/2020**	**Target** (2024)	**Latest known results** (situation on 31/12/2022)
Less than 1 per 1000	Less than 1 per 2000	No incident reported in 2022

Indicator 3: Percentage of staff completing awareness and training plan for security **Source of data:** European Commission		
Baseline **(2020)** **DG DEFIS created on 1/1/2020**	**Target** (2024)	**Latest known results** (situation on 31/12/2022)
0	100%	- EUCI briefing : 80% - RUE training : 34% - SNC training : 40%

Main outputs in 2022:			
Description	**Indicator**	**Target**	**Latest known results** (situation on 31/12/2022)
Update of the internal guide for the handling of SNC and RUE information	Publication of internal guide on DEFIS intracomm	- Mid-2022 - All DEFIS staff	Under finalisation – will be published in Q1 2023
Organise security trainings	- Percentage of management attending awareness raising activities. Percentage of staff attending awareness raising activities.	- Management: 40%. - Staff: 80% - Tailored internal training sessions: 2 in 2022	- Management: 30%. - Staff: 70% - Tailored internal training sessions for all DEFIS staff including management: 1 in 2022 (attendance 62% DEFIS staff)
Staff security clearance	Percentage of staff with personal security clearance or in the process of being security cleared	- all DEFIS staff: 95 %	97.5%
Internal security reports	Circulation of monthly reports	- Management staff - 8 reports	3 reports in 2022 Decision to circulate quarterly report
Installation of electronic door handles in all directorate office spaces	Percentage of equipped doors	- 90 %	95%

ANNEX 11: Implementation through national or international public-sector bodies and bodies governed by private law with a public sector mission (if applicable)

Not applicable.

ANNEX 12: EAMR of the Union Delegations (if applicable)

Not applicable.

ANNEX 13: Decentralised agencies and/or EU Trust Funds (if applicable)

Not applicable.